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Chronic Kidney Disease–Mineral Bone Disorder

Marta Christov | Stuart M. Sprague

CHAPTER OUTLINE

KEY POINTS

- Changes in mineral metabolism parameters occur as early as stage 2 chronic kidney disease (CKD), most commonly with increased serum fibroblast growth factor 23 (FGF-23) concentrations.
- Pi bioavailability depends on the food source, with plant-based sources such as legumes having less bioavailable Pi due to human's inability to digest phytate, the major Pi storage form in plants.
- The commonly used albumin-corrected total calcium values misclassify the true hypercalcemia or hypocalcemia in 20% of patients with CKD, thus whenever possible, ionized calcium concentrations should also be obtained, particularly when starting or modifying therapies for CKD-MBD (mineral bone disease).
- Vascular calcification of the medial layer of elastic arteries is extremely common in patients with CKD, although the pathophysiology may be similar to that of intimal calcifications associated with atherosclerosis.
- There is no definitive noninvasive way (by imaging or bloodwork) to assess bone physiology parameters such as turnover and mineralization.
- The new system for renal osteodystrophy classification focuses on abnormalities in bone turnover, mineralization, and volume, all of which impact bone strength.
- Dual-energy X-ray absorptiometry (DXA) may be used for assessment of fracture risk in patients with CKD, although clinical trials linking changes in DXA to relevant outcomes, including fracture, are lacking.
- There are no randomized controlled trials in CKD or end-stage kidney disease to show that treatment to a specific parathyroid hormone target leads to improved outcomes.
- Patients on dialysis have higher rates of hip fracture than in the general population in all age groups, and have a higher mortality when they experience a fracture.
- Fractures, especially in the appendicular skeleton, are common in kidney transplant recipients.

In persons with healthy kidneys, normal serum levels of phosphorus and calcium are maintained through the interaction of three hormones: parathyroid hormone (PTH); calcitriol [1,25-dihydroxyvitamin D_3 1,25$(OH)_2$D], which is the active metabolite of vitamin D; and fibroblast growth factor 23 (FGF-23). Circulating or soluble klotho also plays a role in mineral homeostasis. These hormones act on four primary target organs: bone, kidney, intestine, and parathyroid glands. The kidneys play a critical role in the regulation of normal serum calcium and phosphorus concentrations and of the three hormones. Thus, derangements in mineral homeostasis are common in patients with chronic kidney disease (CKD). Abnormalities begin early in the course of CKD and are nearly universally observed at a glomerular filtration rate (GFR) less than 30 mL/min. With progression of CKD, the body attempts to maintain normal serum concentrations of calcium and phosphorus by altering the production of calcitriol, PTH, FGF-23, and klotho. Eventually these compensatory responses become unable to maintain normal mineral homeostasis, resulting in (1) altered serum concentrations of calcium, phosphorus, PTH, calcitriol, FGF-23, and klotho; (2) disturbances in bone remodeling and mineralization (often referred to as "renal osteodystrophy") and/or impaired linear growth in children; and (3) extraskeletal calcification in soft tissues and arteries. In 2006, the term chronic kidney disease–mineral and bone disorder (CKD-MBD) was developed to describe this triad of abnormalities in biochemical measures, skeletal abnormalities, and extraskeletal calcification (Table 53.1).[1] These abnormalities that constitute CKD-MBD are interrelated in both the pathophysiology of the disease and the response to treatment. All three components of CKD-MBD are associated with increased risk of fractures, cardiovascular disease, and mortality in patients with CKD stages 4 through 5D. However, to enhance understanding of the complex integration of these abnormalities in CKD, each component is first discussed independently.

Table 53.1 Kidney Disease Improving Global Outcomes Classification of Chronic Kidney Disease–Mineral Bone Disease and Renal Osteodystrophy

Definition of chronic kidney disease–mineral bone disease (CKD-MBD)	A systemic disorder of mineral and bone metabolism due to CKD manifested by one or a combination of the following: Abnormalities of calcium, phosphorus, parathyroid hormone, or vitamin D metabolism Abnormalities in bone turnover, mineralization, volume, linear growth, or strength Vascular or other soft tissue calcification
Definition of renal osteodystrophy	Renal osteodystrophy is an alteration of bone morphology in patients with CKD. It is one measure of the skeletal component of the systemic disorder of CKD-MBD that is quantifiable by histomorphometry of bone biopsy.

From Moe S, Drüeke T, Cunningham J, et al. Definition, evaluation, and classification of renal osteodystrophy: a position statement from Kidney Disease: Improving Global Outcomes (KDIGO). *Kidney Int.* 2006;69:1945–1953.

PATHOPHYSIOLOGY OF CHRONIC KIDNEY DISEASE-MINERAL AND BONE DISORDER

PHOSPHORUS AND CALCIUM HOMEOSTASIS

PHOSPHORUS BALANCE AND HOMEOSTASIS

Inorganic phosphorus is critical for numerous physiologic functions, including skeletal development, mineral metabolism, cell membrane phospholipid content and function, cell signaling, platelet aggregation, and energy transfer through mitochondrial metabolism. Because of the importance of phosphorus in these functions, normal homeostasis maintains serum phosphorus concentrations between 2.5 and 4.5 mg/dL (0.81 and 1.45 mmol/L). Serum concentrations are highest in infants and decrease throughout growth, reaching adult levels in the late teens. Total adult body stores of phosphorus are approximately 700 g, of which 85% is contained in bone in the form of hydroxyapatite $[(Ca)_{10}(PO_4)_6(OH)_2]$. Of the remainder, 14% is intracellular, and only 1% is extracellular. Of this extracellular phosphorus, 70% is organic (phosphate) and contained within phospholipids, and 30% is inorganic. The inorganic fraction is 15% protein bound, and the remaining 85% is either complexed with sodium, magnesium, or calcium or circulates as the free monohydrogen or dihydrogen form. It is this inorganic fraction that is freely circulating and measured. At a pH of 7.4, inorganic phosphates are in a ratio of about 4: 1 HPO_4^{-2} to H_2PO^{-1}. For that reason, the serum phosphorus concentration is usually expressed in mmol/L rather than mEq/L. Thus serum measurements reflect only a minor fraction of total body phosphorus and therefore do not accurately reflect total body stores in the setting of the abnormal homeostasis that occurs in CKD. Furthermore, there is considerable diurnal variation in serum phosphorus concentrations in healthy persons[2] as well as in patients with advanced CKD.[3] The terms phosphorus and phosphate are often used interchangeably, but strictly speaking, "phosphate" means the inorganic freely available form (HPO_4^{-2} and H_2PO^{-1}). However, most laboratories report phosphate, the measurable inorganic component of total body phosphorus, as "phosphorus." For simplicity we use the abbreviation Pi to represent phosphate and/or phosphorus throughout this chapter.

Pi is contained in almost all foods and is generally associated with a food's protein content or phosphate-containing additives. In most commonly ingested foods without additives, the mean Pi content ranges from 9.0 to 14.6 mg per gram of protein, with many foods having up to a 28% higher Pi content because of additives, which are used as preservatives, acidifying agents, acidity buffers, and emulsifying agents.[4] Although the recommended daily allowance for Pi is 800 mg/day, the average American diet contains approximately 1000 to 1400 mg Pi and that amount does not necessarily include inorganic Pi added as a preservative, a common practice.[5] The source of Pi has a significant impact on bioavailability. Pi in the form of preservatives or additives is nearly 100% bioavailable, whereas Pi bound to phytate, as in legumes, is less bioavailable owing to the lack of the enzyme phytase in humans.[6] In studies, the source of Pi directly affects Pi homeostasis.[3] As a result, it is challenging to balance dietary Pi restriction against the need for adequate protein intake in patients with CKD, especially with malnutrition present in up to 50% of patients undergoing dialysis. Pi balance in earlier stages of CKD (stage 3–4) is generally neutral because of the phosphaturic effects of PTH and FGF-23[7]; as these compensatory mechanisms begin to fail and particularly when patients have little to no residual kidney function, however, positive Pi balance likely ensues.

Approximately 60% to 70% of dietary Pi is absorbed by the gastrointestinal tract, predominantly in the small intestine, although transport can occur in all intestinal segments. Pi absorption occurs via passive sodium-independent transport and active sodium-dependent transport. Although it can vary depending on the study and experimental design, active transport is approximately 50% of total transport (Fig. 53.1).[8] Passive transport occurs down electrochemical gradient through paracellular tight junctions; claudins and occludins appear to be involved and to control transport rates and ion specificity.[9] Active absorption occurs via the epithelial brush border type II (solute carrier A34) transporters, specifically the sodium-Pi cotransporter (NaPi-IIb) utilizing energy from the basolateral sodium-potassium ATPase transporter. Complete ablation of the *NaPi-IIb* gene in mice demonstrates that this transporter is responsible for 90% of sodium-dependent transport but only 50% of total intestinal Pi transport.[10] In animals with CKD induced by adenine, ablation of the *NaPi-IIb* gene lowers serum Pi, with additional lowering by the Pi binder sevelamer, suggesting that both active transport and passive transport are important in CKD.[11] The NaPi-IIb transporter is predominantly stimulated by high dietary Pi and calcitriol.[8] In addition, studies suggest that the phosphatonins, matrix extracellular phosphoglycoprotein (MEPE),[12] and FGF-23[13] may play a role in intestinal transport. Tenapanor, an inhibitor of the gut sodium/hydrogen exchanger, also affects Pi transport in the intestine. Tenapanor decreases paracellular absorption by increasing resistance to Pi transport at tight junctions, and it decreases NaPi-IIb expression, thereby decreasing active Pi transport.[14] However, dietary Pi appears the most important regulator of intestinal absorption.

The kidneys are responsible for maintaining Pi balance by excreting the net amount of Pi absorbed (see Chapter 7). Most inorganic Pi is freely filtered by the glomerulus with approximately 70% to 80% reabsorbed in the proximal tubule, which serves as the primary regulated site of the kidney. The remaining 20% to 30% is reabsorbed in the distal tubule. Pi transport across the apical lumen occurs via an active transport

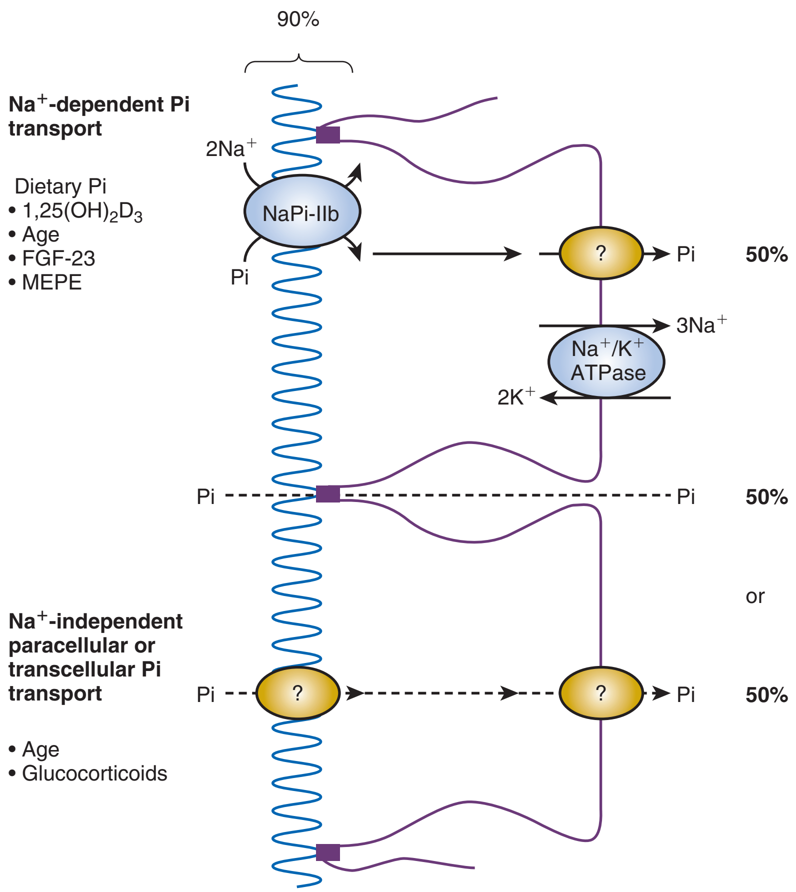


Fig. 53.1 Intestinal phosphate transport. Approximately 50% of phosphate (Pi) transport is sodium (Na^+) dependent, due to active transport, and regulated by a number of factors. The remaining phosphate transport is sodium independent and due to paracellular or transcellular transport. *FGF-23,* Fibroblast growth factor 23; *MEPE,* matrix extracellular phosphoglycoprotein; *Na^+/K^+ ATPase,* sodium-potassium adenosine triphosphatase; *NaPi-IIb,* the sodium Pi cotransporter. (From Lee GJ, Marks J. Intestinal phosphate transport: a therapeutic target in chronic kidney disease and beyond? *Pediatr Nephrol.* 2015;30:363–371. With permission.)

process that is driven by active sodium transport on the basolateral side by the sodium-potassium adenosine triphosphatase (Na^+-K^+-ATPase). The primary transporters on the luminal surface are NaPi-IIa (SLC34A1) and NaPi-IIc (SLC34A3), with a minor component via the type III sodium-dependent Pi cotransporter Pit-2 (SLC20A2). PTH and FGF-23 both downregulate these NaPi transporters, but through different signaling mechanisms. FGF-23 stimulates endocytosis of the transporters after signaling through the FGF receptor (FGFR)–klotho complex, described later. PTH, after binding to PTHR1 receptor, leads to increased cyclic adenosine monophosphate/protein kinase A (cAMP/PKA) signaling with the scaffolding protein Na^+-H^+ exchanger regulatory factors 1 and 3 (NHERF1 and NHERF3) playing an important role. FGF-23 also decreases calcitriol production by reducing the conversion of 25(OH)D to calcitriol and increasing its catabolism. The reduction in calcitriol results in further reduction of intestinal Pi absorption.

CALCIUM BALANCE AND HOMEOSTASIS

Serum calcium concentrations are tightly controlled within a relatively narrow range, usually 8.5 to 10.5 mg/dL (2.1 to 2.6 mmol/L). However, the serum calcium concentration is a poor reflection of overall total body calcium, because serum concentrations are less than 1% of total body calcium. The remainder of total body calcium is stored in bone. Ionized calcium, generally 40% of total serum calcium, is physiologically active, whereas the nonionized calcium is bound to albumin or anions such as citrate, bicarbonate, and Pi. In the presence of hypoalbuminemia, there is a relative increase in the ionized calcium relative to the total calcium; thus total serum calcium measurement may underestimate the physiologically active (ionized) serum calcium. A commonly utilized formula for estimating the ionized calcium from the total calcium value is to add 0.8 mg/dL for every 1-mg decrease in serum albumin below 4 mg/dL. However, in a study in patients with CKD stages 3 to 5 not undergoing dialysis, total calcium concentration and albumin-corrected total calcium values failed to correctly classify 20% of patients as either hypocalcemic or hypercalcemic whose state was documented by ionized calcium measurement.[15] The sensitivity to detect true hypocalcemia or hypercalcemia was only 40% and 21% for total serum calcium, and 36% and 21% for albumin-corrected serum calcium, respectively.[15] The primary causes of this discordance are thought to be albumin, PTH, and pH, although the last has been questioned.[16] Thus, whenever possible, ionized calcium measurement should be utilized. Serum concentrations of ionized calcium are maintained within the normal range by the secretion of PTH, as discussed later.

In normal individuals, the net calcium balance (intake–output) varies with age. Children and young adults are usually in a slightly positive net calcium balance to enhance linear growth; beyond ages 25 to 35 years, when bones stop growing, calcium balance tends to be neutral. Normal individuals have protection against calcium overload by virtue of their ability to increase renal excretion of calcium and reduce intestinal absorption of calcium through the actions of PTH and calcitriol. However, in CKD the ability to maintain normal homeostasis, including a normal serum ionized calcium level and appropriate calcium balance for age, is diminished, and lost completely when kidney function is absent. Two studies in patients with CKD stages late 3 to 4 demonstrate that 1000 mg per day of dietary or calcium supplement/binder leads to near-neutral calcium balance (reviewed in Chapter 19).[7,17]

Calcium absorption across the intestinal epithelium occurs via a vitamin D–dependent, saturable (transcellular) pathway and a vitamin D–independent, nonsaturable (paracellular) pathway. In states of adequate dietary calcium, the paracellular mechanism prevails, but the vitamin D–dependent pathways are critical in calcium-deficient states. The transcellular absorption occurs via three steps: (1) calcium enters from the lumen into the cells via transient receptor potential vanilloid (TRPV) channels, of which TRPV6 is most important in the intestine[18]; (2) the intracellular calcium associates with calbindin-D9K to be "ferried" to the basolateral membrane; and (3) calcium is removed from the enterocytes predominantly via the calcium-ATPase, with the Na^+-Ca^{2+} exchanger playing a minor role. The duodenum is the major site of calcium absorption, although the other segments of the small intestine and the colon also contribute to net calcium absorption. All of the key regulatory components of active calcium transport—TRPV, calbindin, the Ca^{2+}-ATPase (PMCA1b), and the Na^+-Ca^{2+} exchanger (NCX1)—are upregulated by calcitriol.[19] Mice with intestinal knockdown of the vitamin D receptor (VDR) are still able to maintain normal calcium levels because of PTH-induced increased bone resorption; by contrast, global knockdown of VDR leads to hypocalcemia. Hypocalcemia with VDR knockdown can be corrected with either a high-calcium diet (and presumed intestinal paracellular calcium transport) or the administration of calcitriol.[20] Thus bone and kidney can compensate for impaired responsiveness of the intestinal VDR, and diet alone can compensate for a total lack of vitamin D. The 1α-hydroxylase enzyme (CYP27B1) is also located throughout the intestine; to date, however, conversion of 25(OH)D to calcitriol has been identified only in the colonic epithelial cells in inflammation.[21]

The renal transport of calcium is further detailed in Chapter 7. In the kidney, the majority (60%–70%) of calcium is reabsorbed passively in the proximal tubule, a process driven by a transepithelial electrochemical gradient that is generated by sodium and water reabsorption. In the thick ascending limb, another 10% of calcium is reabsorbed via paracellular transport. Calcium-sensing receptor (CaSR) activation in this segment inhibits calcium absorption. This paracellular reabsorption also requires the specific protein paracellin-1, and genetic defects in paracellin-1 lead to a syndrome of hypercalciuria and hypomagnesemia.[22] However, the more regulated aspect of calcium reabsorption occurs via transcellular pathways in the distal convoluted tubule and connecting tubule (Fig. 53.2). The mechanism is similar to intestinal transport: Calcium enters these cells via TRPV5 calcium channels down electrochemical gradients. In the cells, calcium binds with calbindin-D28k and is transported to the basolateral membrane, where calcium is actively reabsorbed by the NCX1 and/or PMCA1b. As in the intestinal epithelial cell, calcitriol upregulates all of these transport proteins.[23] PTH has an indirect effect on renal calcium handling via stimulating the synthesis of calcitriol and increases TRPV5 activity.[24] The enzymatic activity of circulating klotho has been shown to cleave the extracellular domain of TRPV5 channels, keeping these channels at the cell membrane and thereby facilitating calcium reabsorption.[25]

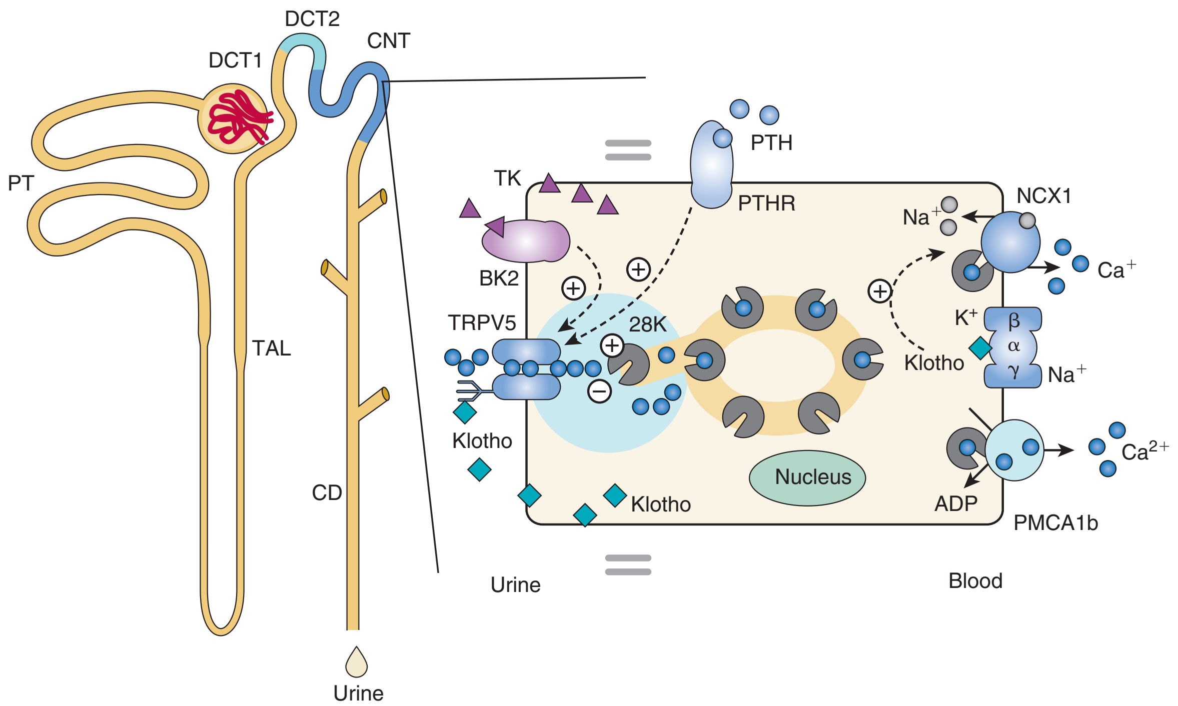


Fig. 53.2 Epithelial calcium active transport. The late part of the distal convoluted tubule *(DCT)* and connecting tubule *(CNT)* play an important role in fine-tuning renal excretion of Ca^{2+}. The epithelial Ca^{2+} channel (TRPV5) is primarily expressed apically in these segments and colocalizes with calbindin-D_{28K} (28K), Na^+/Ca^{2+} exchanger (NCX1), and the plasma membrane adenosine triphosphatase (ATPase; PMCA1b). Upon entry via TRPV5, Ca^{2+} is buffered by 28K and diffuses to the basolateral membrane, where it is released and extruded by a concerted action of NCX1 and PMCA1b. In addition, the basolateral membrane exposes a parathyroid hormone *(PTH)* receptor *(PTHR)* and the Na^+/K^+-ATPase consisting of the α, β, and γ subunits. PTHR activation by PTH stimulates TRPV5 activity, and entered Ca^{2+} can subsequently control the expression level of the Ca^{2+} transporters. At the apical membrane, there is a bradykinin receptor *(BK2)* that is activated by urinary tissue kallikrein *(TK)* to activate TRPV5-mediated Ca^{2+} influx. In the cell, entered Ca^{2+} acts as a negative feedback on channel activity, and 28K plays a regulatory role by association with TRPV5 under low intracellular Ca^{2+} concentrations. Extracellular urinary klotho directly stimulates TRPV5 at the apical membrane by modification of the N-glycan, whereas intracellular klotho enhances Na^+/K^+-ATPase surface expression, which in turn activates NCX1-mediated Ca^{2+} efflux. *ADP,* Adenosine diphosphate; *DCT1,* early part of distal convoluted tubule; *PT,* proximal tubule; *TAL,* thick ascending limb of Henle. (From Boros S, Bindels RJM, Hoenderop JGJ. Active Ca^{2+} reabsorption in the connecting tubule. *Pflügers Arch Eur J Physiol.* 2009;458:99–109.)

CALCIUM-SENSING RECEPTOR

Physiologic studies in animals and humans in the 1980s demonstrated the rapid release of PTH in response to small reductions in blood ionized calcium, lending support to the existence of a CaSR in the parathyroid gland that was subsequently cloned in 1993.[26] The CaSR was shown to belong to the superfamily of G protein–coupled receptors and is a glycosylated protein with a very large extracellular domain, seven membrane–spanning segments, and a relatively large cytoplasmic domain. The primary ligand for the CaSR is Ca^{2+}, but it also senses other divalent and polyvalent cations, including Mg^{2+}, Be^{2+}, La^{3+}, Gd^{3+}, and polyarginine.[27] Extracellular calcium binds to multiple sites, leading to conformational changes that result in activation of phospholipases C, A_2, and D as well as inhibition of cAMP production.[28] Activation of the CaSR stimulates phospholipase C, leading to an increase in inositol 1,4,5-triphosphate (IP_3), which mobilizes intracellular calcium and decreases PTH secretion (Fig. 53.3). By contrast, inactivation of the CaSR reduces intracellular calcium and increases PTH secretion. CaSR messenger RNA (mRNA) is widely expressed in multiple tissues, including organs responsible for CKD-MBD (parathyroid, kidney, thyroid, bone, intestine, vasculature). Studies have demonstrated a diverse role for the CaSR in disease, including in the gastrointestinal

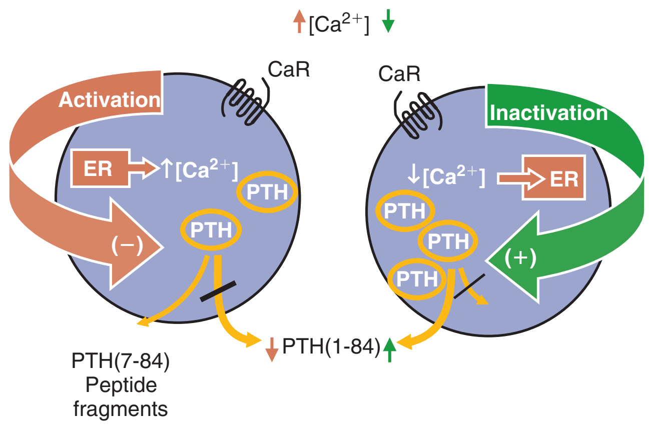


Fig. 53.3 Calcium-sensing receptor *(CaR).* Activation of the CaR by calcium stimulates phospholipase C, leading to increased inositol 1,4,5-triphosphate (IP_3), which mobilizes intracellular calcium and inhibits parathyroid hormone *(PTH)* synthesis. A decrease in serum calcium (Ca^{2+}) inhibits intracellular signaling, leading to increased PTH synthesis and secretion. *ER,* Endoplasmic reticulum. (From Friedman PA, Goodman WG. PTH(1-84)/PTH(7-84): a balance of power. *Am J Physiol Renal Physiol.* 2006;290:F975–F984.)

tract, where it regulates gastrin, glucagon-like peptide-1, acid, and hormone secretion and is involved in taste, gastrointestinal fluid transport, and cell turnover.[29]

CaSR$^{-/-}$ mice die shortly after birth owing to hypercalcemia, hypocalciuria, and hyperparathyroidism. If the *PTH* gene is also ablated, the mice survive and most organs appear histologically normal, with healing of bone mineralization defects but no change in hypocalciuria.[30] However, these CaSR$^{-/-}$/PTH$^{-/-}$ mice demonstrate hypercalcemia in response to oral calcium, infusion of PTH, or administration of calcitriol, whereas the CaSR$^{+/+}$/PTH$^{-/-}$ mice are able to decrease gastrointestinal calcium absorption and increase renal calcium excretion to maintain normal levels of serum calcium.[31,32] These data indicate that CaSR activation corrects hypocalcemia by increasing PTH, whereas in hypercalcemia, the CaSR acts independent of PTH by increasing renal calcium excretion. In uremic animals, the expression of CaSR in the parathyroid gland is downregulated by a high-Pi diet and upregulated by magnesium[33] and calcimimetics. In parathyroid glands from patients with secondary hyperparathyroidism, the expression of the CaSR is downregulated in comparison with expression in nonuremic patients[34] but can be upregulated with the administration of cinacalcet.[35]

The CaSR is expressed throughout the kidney, where it is found in diverse locations and performs multiple physiologic functions: podocyte (cytoskeleton changes), proximal tubule (phosphate reabsorption, calcitriol synthesis, acidification/fluid reabsorption), macula densa (renin secretion), thick ascending loop of Henle (calcium, sodium, potassium, and chloride handling), distal convoluted tubule/connecting tubule (calcium transport), and connecting duct (acid/base, water handling).[36] Diverse functions in the kidney signify how important calcium homeostasis is to normal kidney function and vice versa. Furthermore, many of these functions avoid renal calcium precipitation. Most notably, activation of the luminal CaSR in the collecting duct by elevated urine calcium values leads to urinary acidification and polyuria,[37] a common clinical symptom in hypercalcemia, and prevents calcium-Pi precipitation.

There is some disagreement as to the role of the CaSR in bone and whether or not this role depends on PTH. Clearly bone cells respond to calcium. The CaSR is important in fetal bone development; conditional deletion of the CaSR in early osteoblasts leads to altered bone phenotype, although the results vary depending on the construct used.[38] Studies suggest that the CaSR modulates both bone resorption and formation induced by PTH.[39] In vitro, mesenchymal stem cells appear to require CaSR for differentiation,[40] but in more differentiated cells, calcium channels appear more involved in calcium transport.[41] An initial report found that calcium also regulates FGF-23 synthesis in bone, although this effect does not appear to be mediated via the CaSR.[42] Calcimimetics, allosteric activators of the CaSR, are used to treat secondary hyperparathyroidism as discussed in Chapter 63.

Data also suggest a role for the CaSR in vascular calcification. Immunohistochemical staining demonstrates expression of CaSR on normal human arteries with downregulation in areas of calcification.[43] The CaSR is expressed on cultured vascular smooth muscle cells (VSMCs), and calcimimetics inhibit in vitro calcification.[43,44] The calcimimetic R-568 reverses calcitriol-induced arterial calcification[45] and inhibits proliferation of VSMCs and endothelial cells in the 5/6 nephrectomy rat model.[46] Calcimimetics also retard uremia-enhanced vascular calcification and atherosclerosis in the ApoE$^{-/-}$ mouse[47] and prevent arterial and myocardial calcification in the Cy/+ model of slowly progressive CKD-MBD.[48] Calcimimetics also upregulate a potential local inhibitor of arterial calcification, matrix gla protein.[49] A trial of cinacalcet in patients with end-stage kidney disease (ESKD) treated with calcium-based phosphate binders showed amelioration in calcification relative to placebo (and active vitamin D analogs).[50] These data support a role for the CaSR in all three components of CKD-MBD.

HORMONAL REGULATION OF CHRONIC KIDNEY DISEASE-MINERAL AND BONE DISORDER

PARATHYROID HORMONE

The primary function of PTH is to maintain calcium homeostasis (Fig. 53.4) by (1) increasing bone mineral dissolution, thus releasing calcium and Pi; (2) increasing renal reabsorption of calcium and excretion of Pi; (3) increasing the activity of the renal CYP27B1 enzyme to convert 25(OH)D to calcitriol; and (4) enhancing the gastrointestinal absorption of both

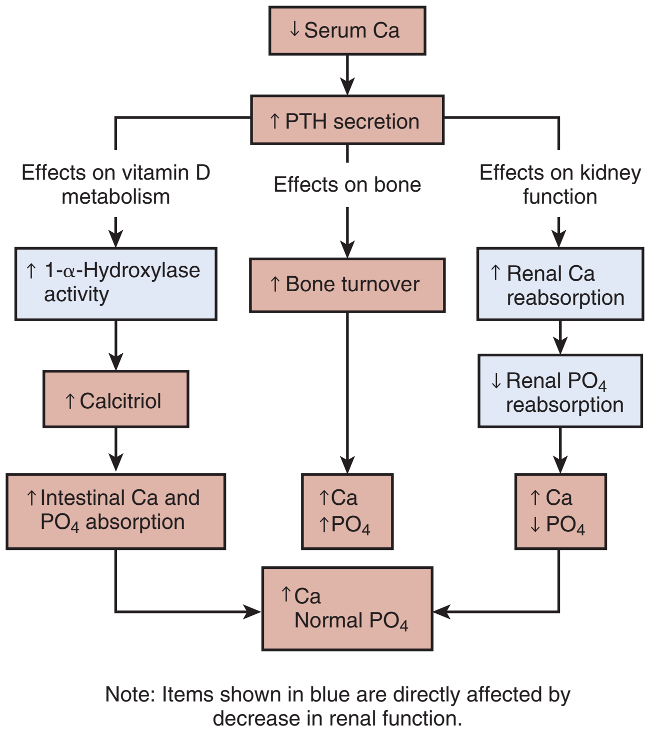


Fig. 53.4 Normalization of serum calcium by multiple actions of parathyroid hormone *(PTH).* Serum levels of ionized calcium *(Ca)* are maintained in the normal range by induction of increases in the secretion of PTH. PTH acts to increase bone resorption, renal calcium reabsorption, and the conversion of 25(OH)D to calcitriol in the kidney, thereby increasing gastrointestinal calcium absorption. The *blue boxes* indicate processes that are abnormal in chronic kidney disease, leading to altered calcium homeostasis. *PO_4,* Phosphate. (From Moe SM. Calcium, phosphorus, and vitamin D metabolism in renal disease and chronic renal failure. In: Kopple JD, Massry SG, eds. *Nutritional Management of Renal Disease.* Philadelphia: Lippincott Williams & Wilkins; 2004:261–285.)

calcium and Pi indirectly through its effects on the synthesis of calcitriol. In healthy persons, the increase in serum PTH concentration in response to hypocalcemia effectively restores serum calcium and maintains serum Pi concentrations. The kidneys are critical to this normal homeostatic response; thus patients with more severe CKD may lose the capacity to maintain calcium homeostasis.

PTH is cleaved to an 84–amino acid protein in the parathyroid gland, where it is stored as fragments in secretory granules for release. Once released, the circulating 1–84–amino acid protein has a half-life of 2 to 4 minutes and is further metabolized in the liver and kidney. PTH secretion occurs in response to hypocalcemia, hyperphosphatemia, and calcitriol deficiency. The extracellular concentration of ionized calcium is the most important determinant of minute-to-minute secretion of PTH from stored secretory granules. The CaSR mediates the rapid response in PTH secretion, which occurs within seconds of changes in ionized calcium concentration. Inactivating mutations of the CaSR have been associated with neonatal severe hyperparathyroidism and benign familial hypocalciuric hypercalcemia. Affected patients have asymptomatic elevations of serum calcium in the presence of nonsuppressed PTH. Activating mutations have been found in patients with autosomal dominant hypocalcemia resulting in inhibition of PTH secretion at relatively lower serum calcium concentrations. PTH is released as both an intact (1–84) protein and carboxy (C)-terminal fragments (often called PTH [7–84]; see Fig. 53.3). C-terminal PTH has the opposite effect on calcium release from bone in animals and cultured calvariae from that of PTH with an intact N terminus.[51] In addition, the C-terminal PTH inhibits apoptosis in osteoblasts, whereas the N-terminal PTH induces apoptosis.[51] Regulators of PTH secretion act by changing the proportion of intact 1–84 and C-terminal fragments.

PTH binds to the PTH1 receptor (PTH1R), which is a member of the G protein–linked seven membrane–spanning receptor family and is widely expressed. PTH-related peptide (PTHrp) shares homology with the first few amino acids of PTH and also binds the PTH1R. Activation of the PTH1R stimulates heterodimeric G proteins G_s (leading to stimulation of cAMP and PKA signaling) and $G_{\alpha q}$ (leading to activation of IP_3 and protein kinase C), ultimately resulting in changes in intracellular calcium.[52] PTH1R activation may vary in response to time exposure, secondary conformational changes after binding, and which cell signaling mechanism is preferentially activated. In general, the effects of PTH are systemic and those of PTHrp are autocrine.

The interrelationship of calcium, Pi, FGF-23, and calcitriol in the development of secondary hyperparathyroidism in CKD is complex and nearly impossible to fully evaluate in humans, because changes in one lead to rapid changes in the others. The response to a decrease in ionized calcium mediated by the CaSR is likely the most potent stimulus for PTH release. Pi increases PTH production by enhancing the stability of PTH mRNA.[53] FGF-23 directly stimulates PTH release in a klotho-independent manner.[54] Calcitriol suppresses PTH release via the VDR to lead to direct suppression of the gene. Other vitamin D compounds that bind to the VDR with lower affinity still reduce PTH release if given in high enough quantities.[55] Although PTH-induced signaling predominantly affects mineral metabolism, there are also many extraskeletal manifestations of PTH excess in CKD. These include encephalopathy, anemia, extraskeletal calcification, peripheral neuropathy, cardiac dysfunction, hyperlipidemia, bone and muscle pain, pruritus, and impotence.[56]

In the kidney, PTH facilitates calcium reabsorption and Pi excretion, as noted earlier. In bone, PTH receptors are located on osteoblasts, with a time-dependent effect. PTH administered long term inhibits osteoblast differentiation and mineralization. By contrast, the administration of PTH to osteoblasts in a pulse rather than a continuous manner stimulates osteoblast proliferation, forming the basis for the administration of PTH as an anabolic therapy for osteoporosis.[57] PTH also interacts with wnt/β-catenin signaling, as discussed in the bone section.

VITAMIN D

Cholesterol is synthesized to 7-dehydrocholesterol, which in turn is metabolized in the skin to vitamin D_3 (Fig. 53.5). This reaction is facilitated by ultraviolet light (ultraviolet B) and higher temperature, and is therefore reduced in individuals with high skin melanin content and inhibited by sunscreen containing sun protection factor 8 or higher. In addition, there are dietary sources of vitamin D_2 (ergocalciferol) and vitamin D_3 (cholecalciferol). The difference between D_2 (plant source) and D_3 (animal source) compounds is the presence of a double bound (D_2) between carbon numbers 22 and 23 in the side chain. Once in the blood, both D_2 and D_3 bind with vitamin D–binding protein (DBP) and are carried to the liver, where they are hydroxylated by CYP27A1 (25-hydroxylase) in an essentially unregulated manner to yield 25(OH)D, often called calcidiol. Once they are converted to calcidiol, there appears to be no difference between the biologic activities of D_2 and D_3. Calcidiol is then converted in the kidney (or other cells) to calcitriol by the action of CYP27B1. This active metabolite is also degraded by other kidney enzymes, 24,25-hydroxylase (CYP24A1), and CYP3A4, providing the primary metabolism of the active compound.

Vitamin D–binding protein is a 58-kDa protein synthesized in the liver. Its serum levels in humans are between 4 and 8 mM, and the protein has a half-life of 3 days. Both the parent vitamin D, 25(OH)D, and calcitriol are carried in the circulation by DBP, but its greater affinity is for 25(OH)D. Targeted gene disruption studies show that DBP-null mice have a marked reduction in both circulating and tissue distributions of calcitriol and yet are normocalcemic, indicating that the primary role of DBP is to maintain stable serum stores of vitamin D metabolites.[58] At the cellular level, both 25(OH)D and calcitriol are endocytosed. Inside the cell, calcitriol can be inactivated by mitochondrial 24-hydroxylase (CYP24A1) or can bind to the VDR in the cytoplasm. Once the VDR-ligand binding has occurred, the VDR translocates to the nucleus, where it heterodimerizes with the retinoid X receptor. This complex binds the vitamin D response element of target genes and recruits transcription factors and corepressors/coactivators that modulate the transcription.[59] These corepressors and coactivators appear to be specific for the ligand, and thus different forms and analogs of vitamin D may produce different effects at each tissue, forming the basis for the pharmacologic development of analogs. Degradation of calcidiol is believed to occur principally in the kidney, from side cleavage and oxidation, to form $24,25(OH)_2D$.[59]

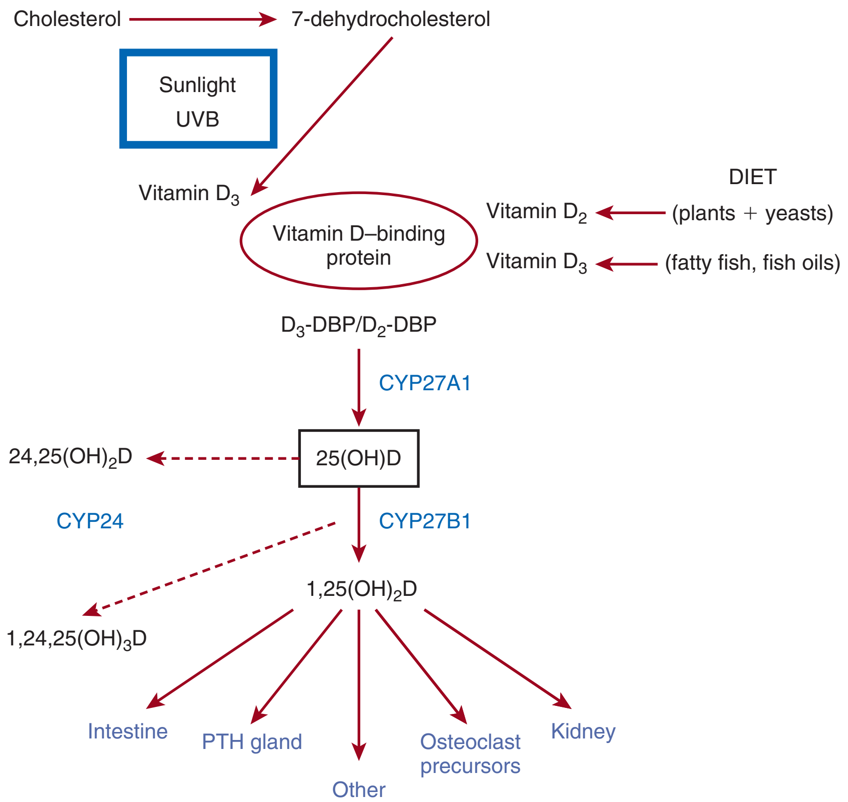


Fig. 53.5 Overview of vitamin D metabolism. Vitamin D is obtained from dietary sources and is metabolized via ultraviolet light (ultraviolet B *[UVB]*) from 7-dehydrocholesterol in the skin. Both sources (diet and skin) of vitamin D_2 and vitamin D_3 bind to vitamin D–binding protein *(DBP)* and circulate to the liver. In the liver, vitamin D is hydroxylated by CYP27A1 *(25-hydroxylase)* to 25(OH)D, commonly referred to as calcidiol. Calcidiol is then further metabolized to calcitriol by the 1α-hydroxylase enzyme *(CYP27B1)* at the level of the kidney. The active metabolite $1,25(OH)_2D$ (calcitriol) acts principally on the target organs of intestine, parathyroid *(PTH)* gland, bone cell precursors, and the kidney. Calcitriol is metabolized to the inert $1,24,25(OH)_3D$ through the action of the 24,25-hydroxylase enzyme *(CYP24)*. Calcidiol is similarly hydroxylated to $24,25(OH)_2D$. (Modified from Moe SM. Renal osteodystrophy. In: Pereira BJG, Sayegh M, Blake P, eds. *Chronic Kidney Disease: Dialysis and Transplantation.* 2nd ed. Philadelphia: Elsevier Saunders; 2004.)

It is generally accepted that the major source of the circulating levels of calcitriol is the kidney. There is evidence that both 25(OH)D and calcitriol have local tissue effects because the VDR, CYP27B1, and CYP24A1 are found in many cells throughout the body.[60] While the kidneys are the predominant site of conversion of 25(OH)D to calcitriol, there is evidence of conversion in other organs, associated with CYP27B1 expression and/or activity in normal and abnormal cells. Other cell types active in conversion of 25(OH)D to calcitriol include osteoblasts, breast epithelial cells (normal and cancerous), prostate gland (normal and cancerous), alveolar and circulating macrophages, pancreatic islet cells, synovial cells, and arterial endothelial cells. Some of these cells may directly take up calcitriol, and others may endocytose the DBP–25(OH)D complex in a megalin-mediated manner (Fig. 53.6), after which the 25(OH)D is hydroxylated by CYP27B1 to act on that specific cell. The presence of CYP24A1 in cells also indicates that the metabolism of calcitriol may be regulated at a cellular level. Circulating concentrations of 25(OH)D are 1000 times higher than those of calcitriol, and thus 25(OH)D exhibits a local, or autocrine/paracrine, effect on many cell types.[60]

Circulating calcitriol mediates its cellular function via nongenomic and genomic mechanisms. Calcitriol facilitates the uptake of calcium in intestinal and renal epithelium by increasing the activity of the voltage-dependent calcium channels TRPV5 and TRPV6. Calcitriol then enhances the transport of calcium through and out of cells, by upregulating the calcium transport protein calbindin (calbindin-D9k in intestine, and calbindin-D28k in kidney) and the basolateral calcium-ATPase as detailed earlier. The CYP27B1 in the kidney is the site of regulation of calcitriol synthesis by numerous other factors, including serum calcium, serum Pi, estrogen, prolactin, growth hormone, FGF-23, and calcitriol itself. Studies show that FGF-23 and inflammatory mediators such as interferon regulate CYP27B1 at nonrenal sites.[61] In vitamin D knockout animals, parathyroid gland hyperplasia is consistently observed despite normalization of serum calcium. However, gland growth can be blunted by exogenous administration of calcitriol even in the absence of VDR, demonstrating a role for calcitriol in regulation of parathyroid gland growth.[62] In the rat, a single small dose of calcitriol decreases PTH secretion by nearly 100%. Studies in the 1970s demonstrated that oral calcitriol, but not the precursor hormone vitamin D_3, suppressed PTH in patients undergoing dialysis, leading to widespread use of calcitriol or its analogs in the management of secondary hyperparathyroidism. However, later studies in animals have demonstrated efficacy

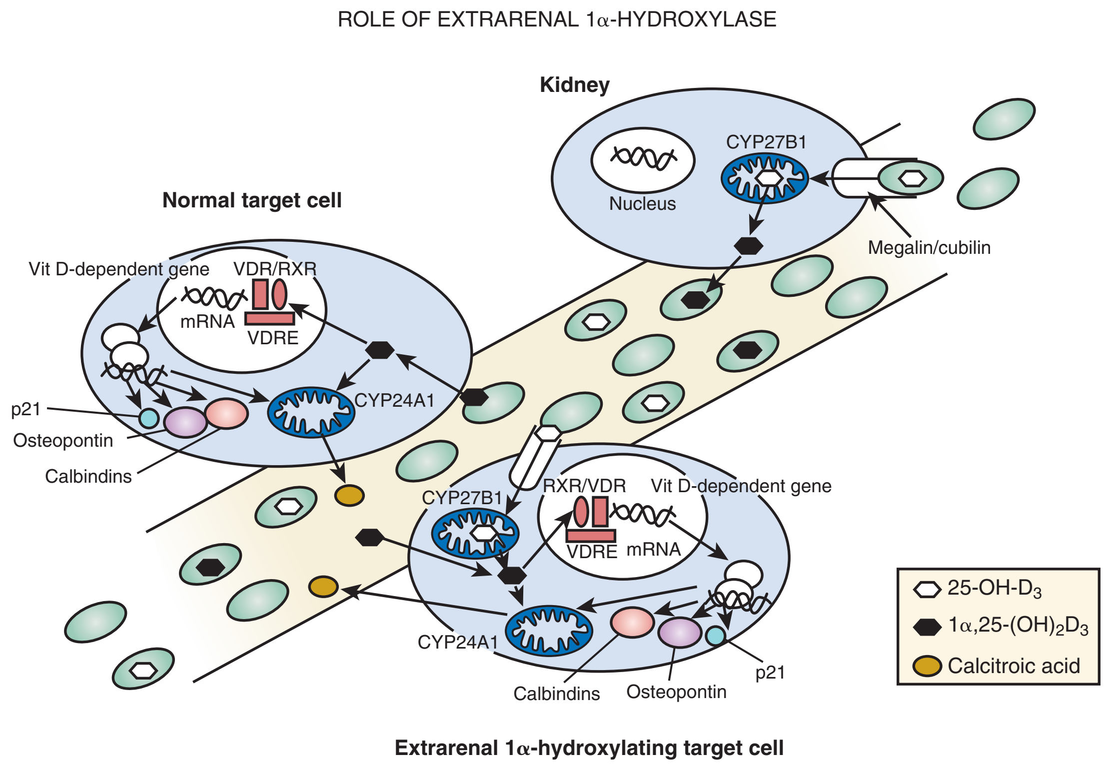


Fig. 53.6 Concept of the role of the extrarenal 1α-hydroxylase. The metabolism of vitamin D in the context of the cells involved is shown. *Upper right,* Proximal tubular cell showing the key elements in the uptake of $25(OH)D_3$ and its conversion to $1\alpha,25(OH)_2D_3$. Megalin/cubilin are cell surface receptors that execute endocytosis of the vitamin D–binding protein (DBP)–$25(OH)D_3$ complex, and CYP27B1 is the main component of the 1α-hydroxylase, responsible for making $1\alpha,25(OH)_2D_3$. *Middle left,* Simple target cell that takes up 1α,25(OH, the vitamin D response element $[VDRE])_2D_3$ as the free ligand originally ferried to the target cell bound to DBP. The picture shows the key elements of the transcriptional machinery as well as some representative gene products, including the cell division protein p21, the bone matrix protein osteopontin, the calcium transport protein calbindin, and the autoregulatory protein CYP24A1. *Lower right,* Target cell expressing extrarenal 1α-hydroxylase, which possesses megalin/cubilin machinery to take up the DBP–$25(OH)D_3$ complex and also expresses CYP27B1, enabling it to make $1\alpha,25(OH)_2D_3$ intracellularly and also to respond in a likewise manner to the simple target cell because it also possesses the vitamin D receptor *(VDR)* and other transcriptional machinery. The expectation is that cells involved in cell differentiation or in control of cell division require higher concentrations of $1\alpha,25(OH)_2D_3$ in order to modulate a different set of genes, and that the CYP27B1 boosts local production to augment "circulating" $1\alpha,25(OH)_2D_3$ arriving from the kidney in the bloodstream. With normal physiologic processes, locally produced $1\alpha,25(OH)_2D_3$ would not enter the general circulation, although in pathologic conditions (e.g., sarcoidosis) it might. At this time, it is not clear how many cell types can be considered simple target cells and how many possess the CYP27B1 and megalin/cubilin to allow for local production of hormone. *mRNA,* Messenger RNA; *RXR,* retinoid X receptor. (From Jones G. Expanding role for vitamin D in chronic kidney disease: importance of blood 25-OH-D levels and extra-renal 1α-hydroxylase in the classical and nonclassical actions of 1α,25-dihydroxyvitamin D_3. *Semin Dial.* 2007;20:316–324. With permission.)

of 25(OH)D in suppression of PTH, but the doses required are much greater than for calcitriol. Studies in humans have shown efficacy of 25(OH)D in suppressing PTH in patients with advanced CKD, but direct comparison studies are lacking.[63]

Calcitriol has multiple effects on many cells that are important in bone remodeling; therefore it is not surprising that bone defects are well described in vitamin D–deficient states. However, the direct effects of the vitamin D system on bone have been difficult to differentiate from the secondary effects of hypocalcemia and hyperparathyroidism in vitamin D–deficient models. Transgenic animals, including 1α-hydroxylase$^{-/-}$/VDR$^{-/-}$ and 1α-hydroxylase$^{-/-}$/VDR$^{-/-}$, have impaired bone mineralization. In these animals, mineralization can be corrected with normalization of serum calcium; even exogenous calcitriol does not fully correct mineralization in the 1α-hydroxylase$^{-/-}$ animals unless serum calcium concentrations are also restored. Studies evaluating bone remodeling also demonstrate an important role for the calcitriol/VDR system. If hypocalcemia is not corrected (leading to secondary hyperparathyroidism), there is increased osteoblast activity and bone formation from the anabolic effects of PTH. The activation of osteoclasts by PTH is blunted, suggesting a synergistic effect of vitamin D and PTH. Supporting this suggestion is the finding that when serum calcium concentrations are corrected by "rescue" diets and secondary hyperparathyroidism is prevented, osteoblast numbers, mineralization activity, and bone volume are still reduced. Comparison studies of 1α-hydroxylase$^{-/-}$ and PTH$^{-/-}$ mice demonstrate a predominant role for PTH in appositional bone growth and for vitamin D in endochondral bone formation. Thus the calcitriol/VDR system has anabolic bone effects that are necessary for bone formation and are supplemental to the effect of PTH.[64]

FGF-23 AND KLOTHO

"Phosphatonins" are circulating factors that regulate urinary Pi excretion. Two main phosphatonins have been described:

FGF-23 and MEPE. Various forms of rickets have now all been found to be due to abnormalities in FGF-23. Autosomal dominant hypophosphatemic rickets is rare and is associated with a mutation that limits normal degradation of FGF-23. Autosomal recessive hypophosphatemic rickets is also rare and is due to a mutation in dentin matrix protein (DMP), a locally produced inhibitor of FGF-23. X-linked hypophosphatemic rickets is the most common form of rickets due to a mutation in *PHEX* (phosphate-regulating gene with homologies to endopeptidases located on the X chromosome). Mutations in *PHEX* have been found to have deficient degradation of FGF-23 in the osteocyte, leading to inappropriately high serum concentrations of FGF-23.[65] Thus what previously was thought to be disorders of different etiologies are now all linked to FGF-23.

FGF-23 is a 251–amino acid hormone predominantly produced from bone cells (osteocytes and osteoblasts) during active bone remodeling, but its mRNA is also found in the heart, liver, thyroid/parathyroid, intestine, and skeletal muscle.[66] FGF-23 production in the osteocyte is stimulated by PTH[67] and calcitriol.[68] Elevated Pi or Pi load and hypercalcemia may also stimulate FGF-23 but this stimulation appears to be indirect. In the osteocyte, both DMP1 and PHEX protein degrade FGF-23 such that mutations in their corresponding genes lead to excess FGF-23.[65] In turn, calcitriol increases PHEX and FGF-23 inhibits calcitriol, completing a feedback loop. Regulation of NaPi-IIa by FGF-23 is independent of PTH; FGF-23 also inhibits the conversion of 25(OH)D to calcitriol by inhibition of CYP27B1 in the renal tubules[69] and at extrarenal sites,[61] and increases catabolism of calcitriol by activation of CYP24,[69] leading to hypophosphatemia and inappropriately normal or low serum calcitriol concentrations. An overview of the FGF–klotho axis is shown in Fig. 53.7.

FGF-23 is a member of a diverse family of 18 FGFs that bind to one of four receptors (FGFRs) via a heparan sulfate cofactor– or klotho coreceptor–dependent manner, leading to diverse biologic effects.[70] Identification of klotho as a coreceptor for FGF-23 was due to nearly identical phenotypes of the knockout mice, including hyperphosphatemia, hypercalcemia, and excess calcitriol levels associated with early mortality, growth retardation, vascular calcification, cardiac hypertrophy, and osteopenia.[71] Klotho was originally identified as an aging suppressor gene. α-Klotho is expressed in the kidney and parathyroid gland and forms complexes with FGFR1 and FGFR4 to enhance FGF-23 signaling. β-Klotho is expressed in the liver and fat, forms complexes with FGFR1 and FGFR4, and supports FGF-15/19 and FGF-21 signaling. γ-Klotho increases FGF-19 activity and is expressed in the eye, fat, and kidney. All three klothos are transmembrane proteins, with short intracellular domains and large extracellular domains that have β-glucosidase cleavage sites. In animals, α-klotho expressed in the distal tubule can be cleaved to release the extracellular domain into the circulation.

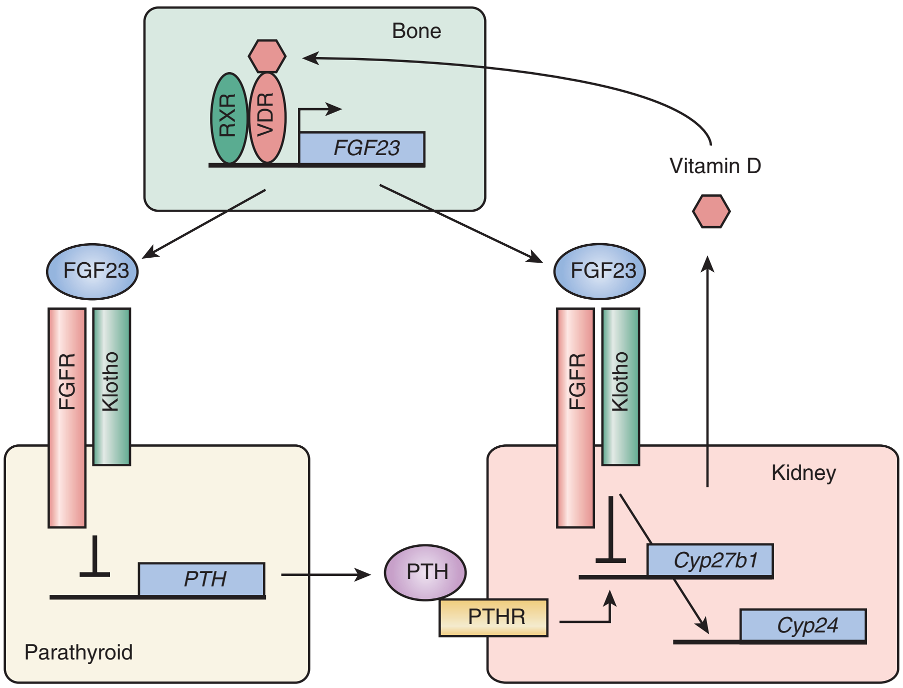


Fig. 53.7 The bone–kidney–parathyroid endocrine axes mediated by fibroblast growth factor 23 *(FGF-23)* and klotho. Active form of vitamin D (calcitriol) binds to vitamin D receptor *(VDR)* in the bone (osteocytes). The ligand-bound VDR forms a heterodimer with a nuclear receptor *(RXR)* and transactivates expression of the FGF-23 gene. FGF-23 secreted from bone acts on the klotho–FGF receptor *(FGFR)* complex expressed in the kidney (the bone-kidney axis) and parathyroid gland (the bone-parathyroid axis). In the kidney, FGF-23 suppresses synthesis of active vitamin D by downregulating expression of the *Cyp27b1* gene and promotes its inactivation by upregulating expression of the *Cyp24* gene, thereby closing a negative feedback loop for vitamin D homeostasis. In the parathyroid gland, FGF-23 suppresses production and secretion of parathyroid hormone *(PTH)*. PTH binds to the PTH receptor *(PTHR)* expressed on renal tubular cells, leading to upregulation of *Cyp27b1* gene expression. Thus suppression of PTH by FGF-23 reduces expression of the *Cyp27b1* gene and serum levels of calcitriol. This step closes another long negative feedback loop for vitamin D homeostasis. (From Kuro-o M. Overview of the FGF23-Klotho axis. *Pediatr Nephrol.* 2010;25:583–590. With permission.)

Further, alternative splicing of the α-klotho gene produces only the extracellular domain that is released from cells and is called circulating or soluble klotho.[72] The soluble α-klotho acts as a coreceptor at nonrenal sites, and prevents FGF-23-induced cardiac hypertrophy.[71] The kidney regulates both renal production of soluble klotho and renal excretion.[72]

In the kidney, tissue klotho is downregulated[72] and FGF-23 is upregulated early in the course of CKD.[73] Klotho is shed from the distal convoluted tubule to serve as a coreceptor with FGF-23 on the proximal tubule, where it inhibits reabsorption of Pi that is regulated by NaPi-IIa, NaPi-IIc, and Pit-1 (similar to PTH but via different signaling mechanisms), and suppresses CYP27B1 to inhibit calcitriol production (opposite of PTH).[74] α-Klotho also increases calcium reabsorption by stimulating TRPV5,[75] decreases potassium excretion through effects on the ROMK1 (renal outer medullary potassium 1) channel,[76] protects against kidney injury and fibrosis,[77] and decreases insulin resistance.[74] Table 53.2 demonstrates the parallel changes in klotho deficiency and CKD.

Table 53.2 Comparison of Phenotypes of Klotho Deficiency and Chronic Kidney Disease

	Klotho Deficiency	Chronic Kidney Disease
Blood Chemistry:		
Phosphate	↑↑↑↑	↑ or ↑↑↑[a]
Calcium	↑	↔ or ↓↓
Creatinine	↑	↑↑↑
Calcitriol	↑↑↑	↓↓↓
Parathyroid hormone	↔ or ↓	↑↑
Fibroblast growth factor 23	↑↑↑	↑↑
Klotho	↓↓↓ or disappears	↓↓ at ESKD[b]
Gross Phenotypes:		
Body weight	↓↓↓	↓↓
Growth retardation	↓↓↓↓	↓↓ in children
Physical activity	↓↓↓	↓
Fertility	↓↓↓↓	↓↓
Life span	↓↓↓↓	↓↓
Cardiovascular Disease:		
Cardiac hypertrophy	↑↑	↑↑↑
Cardiac fibrosis	↑↑	↑↑↑
Vascular calcification	↑↑↑↑	↑↑
Atherosclerosis	↑↑	↑↑↑
Blood pressure	↑	↑↑↑↑
Hematocrit levels	↓	↓↓↓↓
Bone disease	↓↓↓	↓↓↓

[a]During early chronic kidney disease, blood phosphate level is in the normal range.
[b]Blood klotho may be increased in early chronic kidney disease.
ESKD, End-stage kidney disease; ↓, decreases; ↑, increases; ↔, unchanged.
Modified from Hu MC, Kuro-o M, Moe OW. Renal and extrarenal actions of Klotho. *Semin Nephrol.* 2013;33:118–129.

In addition to klotho's role in mineral metabolism, klotho and FGF-23 are implicated in cardiovascular disease. FGF-23 can induce cardiac hypertrophy and increases intracellular calcium in a klotho-independent manner.[71,78,79] Klotho is localized in the heart at the sinoatrial node, and klotho deficiency may lead to arrhythmias.[80] Klotho suppresses cardiomyocyte apoptosis[81] and downregulates TRPC6 (transient receptor potential cation 6) calcium channel in the cardiomyocyte.[82] Both klotho and FGFR1 and FGFR3, but not FGF-23 and FGFR4, are expressed in human arteries.[83,84] In human arteries from patients with CKD, klotho and FGFR1 and FGFR3 are downregulated in the presence of calcification. VDR activators upregulate klotho, leading to an anticalcific effect on FGF-23-induced calcification.[84] Decreased klotho impairs endothelial function.[85] Activation of the renin–angiotensin–aldosterone system reduces renal klotho expression.[86] The pace of discovery of the systemic effects of FGF-23 and klotho has revolutionized our understanding of CKD-MBD.

BONE BIOLOGY

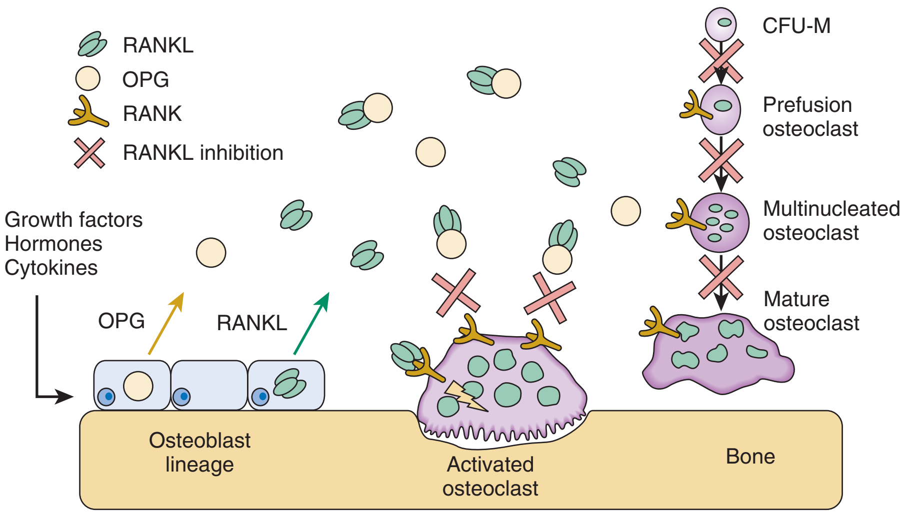
The majority of the total body stores of calcium and Pi is located in bone and therefore bone plays an integral role in homeostasis. Trabecular (cancellous) bone is located predominantly in the epiphyses of the long bones, is 15% to 25% calcified, and serves a metabolic function, with a relatively short turnover time as shown by calcium[45] studies. By contrast, cortical (compact) bone is located in the shafts of long bones and is 80% to 90% calcified. This bone serves primarily a protective and mechanical function and has a calcium turnover time of months. Bone consists principally (90%) of highly organized cross-linked fibers of type I collagen; the remainder consists of proteoglycans and "noncollagen" proteins such as osteopontin, osteocalcin, osteonectin, and alkaline phosphatase. Hydroxyapatite—$Ca_{10}(PO_4)_6(OH)_2$—is the primary bone crystal.

The cellular components of bone are cartilage cells, which are critical to bone development; osteoblasts, which are the bone-forming cells; and osteoclasts, which are the bone-resorbing cells. Osteoblasts are derived from progenitor mesenchymal cells located in the bone marrow. They are then induced to become osteoprogenitor cells, then endosteal or periosteal progenitor cells, then mature osteoblasts. The control of this differentiation pathway is complicated and involves integration of circulating hormones, locally produced factors from the mesenchymal–hematopoietic cell niche, and transcription factors. Once bone formation is complete, osteoblasts may undergo apoptosis or may become quiescent cells trapped within the mineralized bone in the form of osteocytes.[87] The osteocytes are interconnected through a series of canaliculi and serve as mechanoreceptors. Osteocytes detect and respond to mechanical loading and initiate bone remodeling by regulating local osteoclastogenesis via paracrine signals. Osteoclasts are derived from hematopoietic precursor cells that differentiate and are signaled to arrive at a certain place in the bone through the osteoprotegerin (OPG)/RANKL (receptor activator of nuclear factor κB ligand) system detailed later. Once there, they fuse to form the multinucleated cells known as osteoclasts, which become highly polarized, reabsorbing bone through the release of derivative enzymes. These cells move along a resorption surface via changes in the cytoskeleton. PTH, cytokines, and calcitriol are all important in inducing the fusion of the committed osteoclast precursors.

The control of bone remodeling is highly complex, but appears to occur in very distinct phases, as follows: (1) osteoclast recruitment and activation, (2) osteoclast resorption, (3) preosteoblast migration and differentiation, (4) osteoblast deposition of matrix (osteoid or unmineralized bone), (5) mineralization, and (6) quiescence. At any one time, less than 15% to 20% of the bone surface is undergoing remodeling, and this process in a single bone remodeling unit can take 3 to 6 months.[88] How or why a certain segment of bone undergoes a remodeling cycle is not completely clear. The three main systems that interact to regulate remodeling are OPG/RANKL, sclerostin/Wnt/β-catenin, and PTH/PTHR1, which are discussed separately.

The identification of the OPG and RANK system in the 1980s sheds new light on the control of osteoclast function and the long observed coupling of osteoblasts and osteoclasts. RANK is located on osteoclasts, and RANK ligand (RANKL) is secreted by osteoblasts. Osteoblasts also synthesize the decoy protein OPG, which can bind to RANK ligand and inhibit the subsequent binding of RANKL to RANK on osteoclasts, thus inhibiting bone resorption (Fig. 53.8). Alternatively, if OPG production is decreased, RANKL can bind with RANK on osteoclasts and induce osteoclastic bone resorption. This control system is regulated by nearly every cytokine and hormone thought important in bone remodeling, including PTH, calcitriol, estrogen, glucocorticoids, interleukins, prostaglandins, and members of the transforming growth factor-β superfamily of cytokines.[89] OPG has been successful in preventing bone resorption in models of osteoporosis as well as hormone- and cytokine-induced bone resorption,[89] and denosumab, an anti-RANKL antibody, is an approved anabolic drug for the treatment of osteoporosis.[90] Interestingly, abnormalities in the OPG/RANKL system have been found in kidney disease,[91] and early animal models suggest that treatment with OPG may have a protective role in hyperparathyroid bone disease.[92] Initial studies in patients receiving dialysis have demonstrated hypocalcemia as a severe adverse effect of denosumab.[93] More information is required to understand how this system regulates bone remodeling in the context of CKD.

Genetic defects in the gene *SOST* have been identified in rare bone disorders. Sclerostin, the protein product of this gene, binds to low-density lipoprotein receptor–related proteins 5 and 6 (LRP5/LRP6) on the osteocyte to competitively inhibit the binding of the protein wnt (Fig. 53.9). Normally, wnt binding to LRP5/LRP6 leads to stabilization of β-catenin (canonical pathway) and regulation of normal bone accrual via osteoblast differentiation. In the presence of sclerostin, the β-catenin is degraded and mesenchymal stem cell differentiation to mature bone cells is inhibited. In animal models, sclerostin deletion enhances bone accrual,[94] and in early human trials, treatment with an antibody to sclerostin was found to be anabolic.[95,96] Given that sclerostin concentrations are elevated in the blood and bone of patients with CKD[97] and bone in animals with CKD,[98] the anabolic agent antisclerostin antibody may be efficacious in the treatment of renal osteodystrophy. However, initial studies in animals found that antisclerostin antibody was not efficacious in the setting of elevated PTH, although it did improve bone



RANKL inhibition = x

Fig. 53.8 Role of osteoprotegerin *(OPG)*/receptor activator of nuclear factor kappaB ligand *(RANKL)* in bone remodeling. Mechanisms of action for OPG, RANKL, and RANK (receptor activator of nuclear factor kappaB) are depicted in this diagram. RANKL is produced by osteoblasts, bone marrow stromal cells, and other cells under the control of various proresorptive growth factors, hormones, and cytokines. Osteoblasts and stromal cells produce OPG, which binds to and thereby inactivates RANKL. The major binding complex is likely to be a single OPG homodimer interacting with high affinity with a single RANKL homotrimer. In the absence of OPG, RANKL activates its receptor, RANK, found on osteoclasts and preosteoclast precursors. RANK–RANKL interactions lead to preosteoclast recruitment, fusion into multinucleated osteoclasts, osteoclast activation, and osteoclast survival. Each of these RANK-mediated responses can be fully inhibited by OPG. *CFU-M,* Macrophage colony-forming unit. (From Kearns AE, Khosla S, Kostenuik PJ. Receptor activator of nuclear factor kappaB ligand and osteoprotegerin regulation of bone remodeling in health and disease. *Endocr Rev.* 2008;29:155–192.)

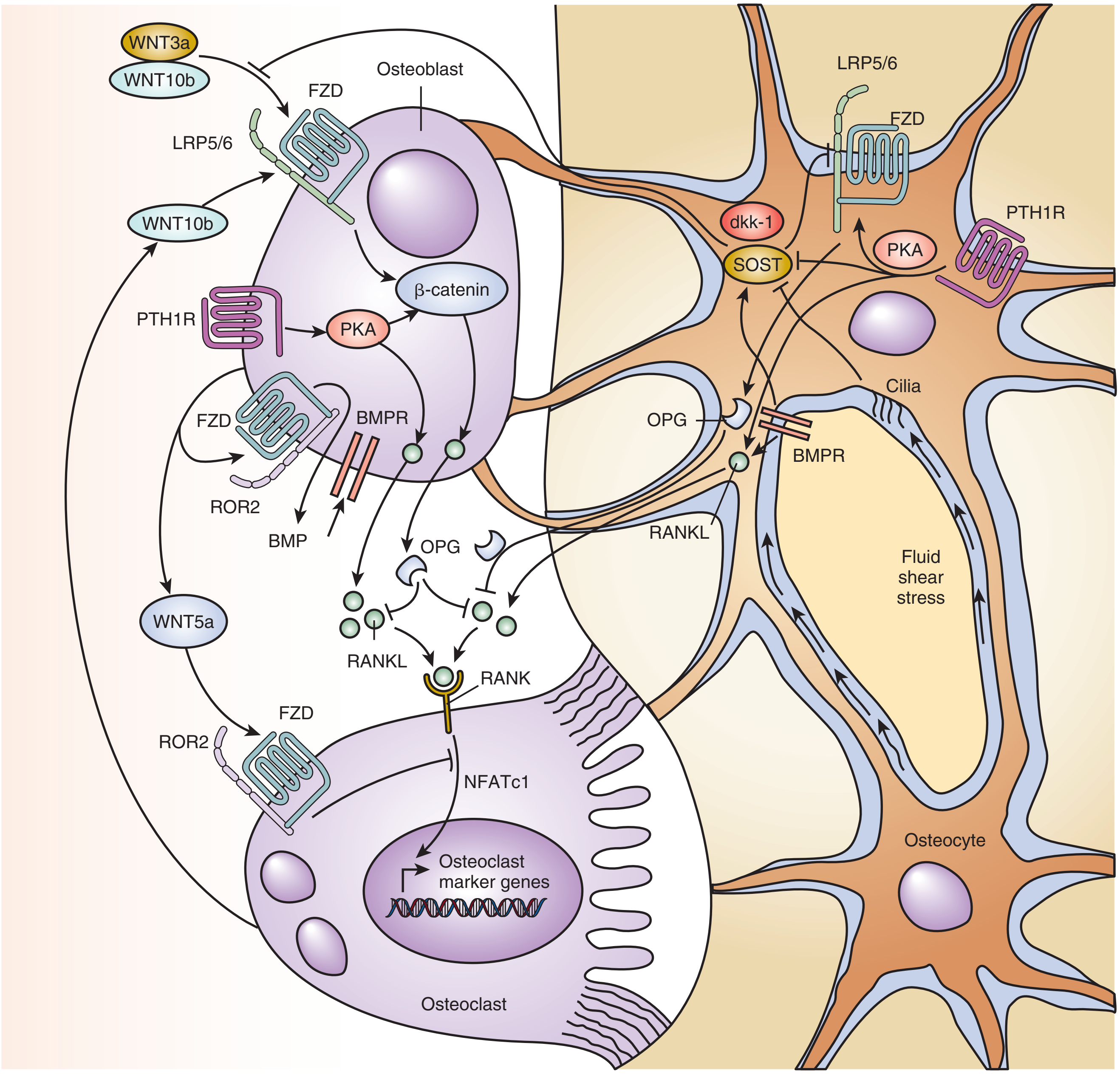


Fig. 53.9 Parathyroid hormone *(PTH)* and β-catenin signaling in bone remodeling. Osteocytes control bone formation through the secretion of the WNT antagonists sclerostin *(SOST)* and Dickkopf WNT signaling pathway inhibitor 1 *(dkk-1),* the expression of which is regulated by mechanosignals and by signaling of PTH and bone morphogenetic protein *(BMP)*. PTH represses expression of these antagonists, whereas BMP signaling, which is mediated by BMP receptor 1A *(BMPR1A),* induces their expression. Moreover, WNT signaling in osteocytes controls the production of osteoprotegerin *(OPG)*, which is the decoy receptor for the key osteoclast differentiation factor RANKL (receptor activator of nuclear factor kappaB ligand). Osteoblast-expressed WNT5a stimulates differentiation of osteoclast precursors as a result of binding to the FZD–ROR2 (frizzled and receptor tyrosine kinase–like orphan receptor 2) receptor complex. In a feedback loop for bone remodeling, osteoclasts stimulate the local differentiation of osteoblasts at the end of the resorption phase by secreting WNT ligands. In addition, activation of parathyroid hormone 1 receptor *(PTH1R)*–mediated signaling in osteoblasts and osteocytes leads to stabilization of β-catenin and, thus, activation of WNT signaling. *LRP5/6,* Low-density lipoprotein (LDL) receptor–related proteins 5 and 6; *PKA,* protein kinase signaling. (From Baron R, Kneissel M. WNT signaling in bone homeostasis and disease: from human mutations to treatments. *Nat Med.* 2013;19:179–192. With permission.)

volume when PTH was suppressed.[99] Dickkopf-related protein 1 (dkk-1) also inhibits wnt binding to LRP5/LRP6, and an antibody to this circulating inhibitor of wnt signaling improved bone remodeling in a model of early CKD.[100] In osteocytes, PTH directly suppresses sclerostin and dkk-1 secretion[67,101] and thus inhibits the production of circulating inhibitors of wnt signaling.

In bone, PTH binds to its receptor, PTH1R, and activates β-catenin signaling via multiple mechanisms (Fig. 53.9): (1) direct activation through cAMP signaling; (2) indirect activation via osteoclast activation, which then increases β-catenin activity in osteoblasts; and (3) by binding to LRP6 to activate LRP5/LRP6 signaling even in the absence of wnt ligands.[94] Thus PTH can activate β-catenin through non–wnt-mediated pathways and through pathways not regulated by sclerostin or dkk-1. There are also differences in responses to continuous and intermittent PTH exposures. Mice expressing a constitutively active PTH1R or animals receiving continuous infusion of PTH(1–84) (analogous to secondary hyperparathyroidism) also have wnt-dependent remodeling with increased osteoclast bone resorption via the OPG/RANKL system, leading to osteoblast activation and β-catenin activation.[102,103] Hyperphosphatemia also activates β-catenin signaling.[104] Thus in CKD with hyperphosphatemia and secondary hyperparathyroidism, there is activation of β-catenin by PTH-mediated inhibition of circulating inhibitors of wnt signaling (sclerostin and dkk-1), PTH-mediated effects independent of wnt signaling, and phosphorus-mediated effects, all leading to enhanced mesenchymal differentiation to osteoblasts, increased RANKL-induced osteoclast activation, and increased bone resorption.

PATHOPHYSIOLOGY OF VASCULAR CALCIFICATION

Vascular disease may be due to a variety of different pathologic processes in different arterial segments, all of which can be calcified. Atherosclerotic disease is characterized by fibro-fatty plaque formation, and on the basis of autopsy data and animal models, calcification had been thought to occur late in the disease course. These plaques can protrude into the arterial lumen, leading to a filling defect on angiography (Fig. 53.10A). However, advances in imaging, especially intravascular ultrasonography, have demonstrated that atherosclerosis can also be a circumferential lesion (without an obstructed lumen) with calcification earlier in the course of the disease.[105] The medial layer may also be affected in arteriosclerosis, leading to thickening commonly found in elastic arteries (Fig. 53.10B). In addition to the larger elastic arteries, smaller elastic arteries may be affected by medial thickening and calcification, classically described as Mönckeberg calcification, or medial calcinosis. This condition is more common in advanced age, and in patients with diabetes mellitus and/or CKD, and is associated with all-cause and cardiovascular mortality in patients with diabetes with and without CKD, as well as in patients with CKD with or without diabetes.

Although initially believed to be related to spontaneous precipitation in the setting of high serum concentrations of calcium and Pi, vascular calcification is now known to be a tightly regulated process that resembles mineralization in bone, a process kept "in check" through the actions of inhibitors of calcification. The current hypothesis accepted by most investigators is that VSMCs dedifferentiate or transform to osteocyte/chondrocyte-like cells (Fig. 53.11). These cells then lay down an extracellular matrix of collagen and noncollagenous proteins and make matrix vesicles that attach to the extracellular matrix to initiate and propagate mineralization. This process is regulated by the cells, the extracellular matrix proteins, and inhibitors that may act locally or systemically. In advanced CKD, there is abnormal bone remodeling and reduced renal clearance of phosphate, generating a positive calcium and Pi balance that "feeds" the mineral composition of matrix vesicles and augments the ability of existing calcification to expand. The evidence for each of these steps is discussed in the following sections.

CELLULAR TRANSFORMATION

VSMCs, osteoblasts, chondrocytes, and adipocytes differentiate from mesenchymal precursors with normal differentiation

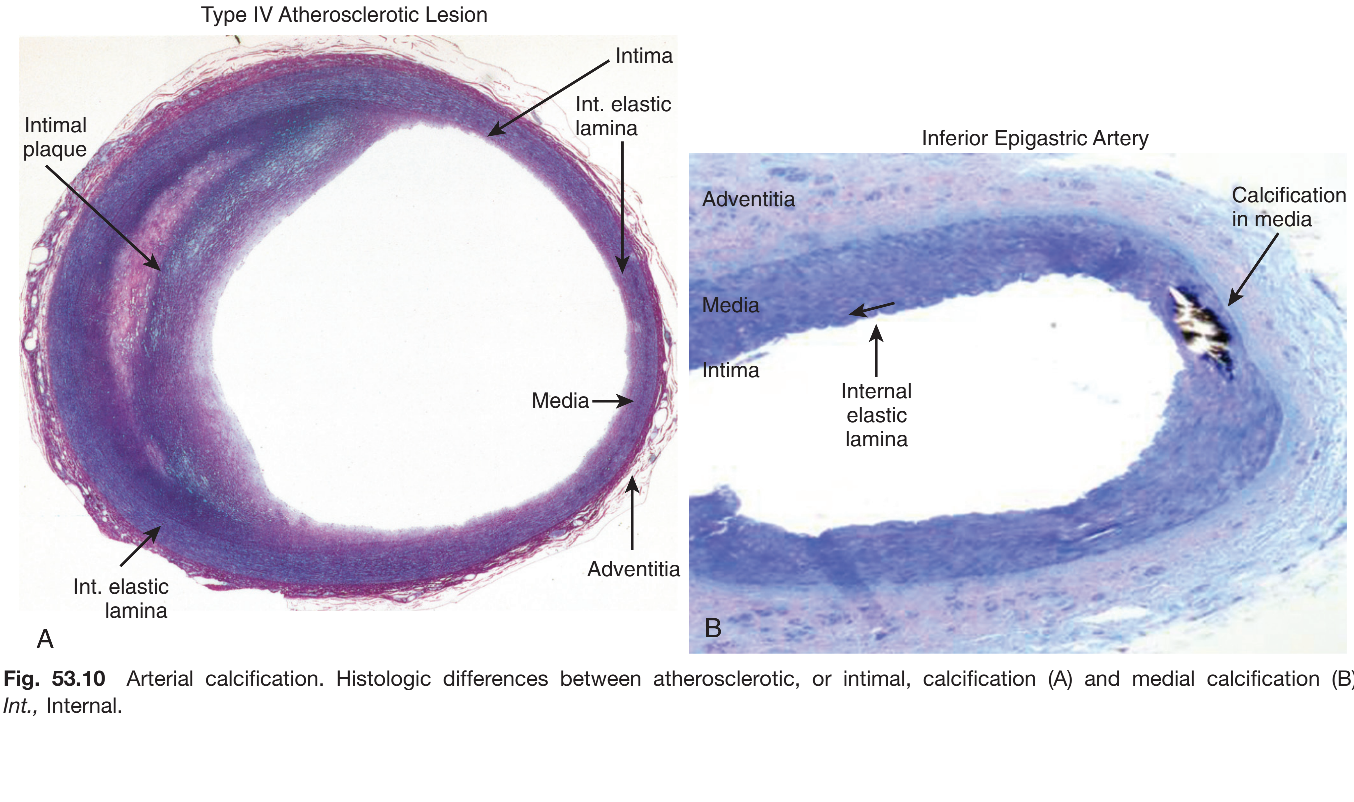


Fig. 53.10 Arterial calcification. Histologic differences between atherosclerotic, or intimal, calcification (A) and medial calcification (B). *Int.,* Internal.

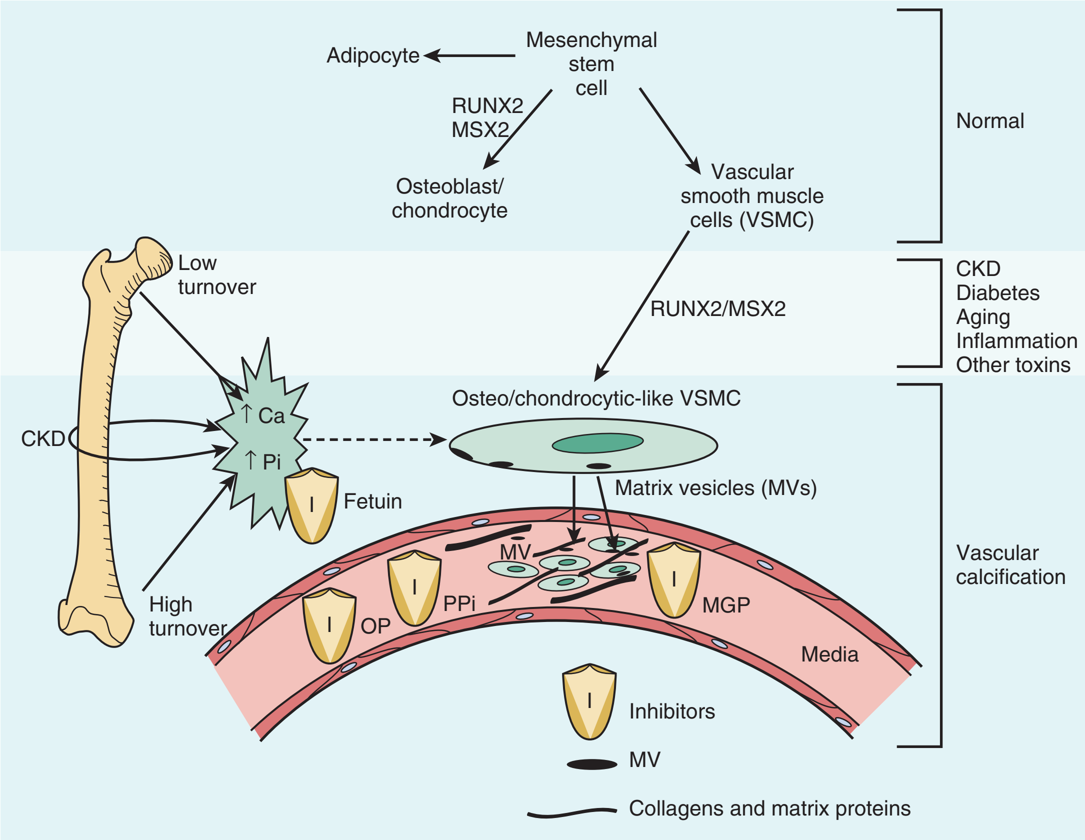


Fig. 53.11 Overview of the pathophysiology of vascular calcification. Normally, mesenchymal stem cells differentiate to adipocytes, osteoblasts, chondrocytes, and vascular smooth muscle cells *(VSMCs)*. In the setting of chronic kidney disease *(CKD)*, diabetes, aging, inflammation, and the presence of multiple other toxins, these VSMCs can dedifferentiate or transform to chondrocyte/osteoblast-like cells by upregulation of transcription factors such as runt-related transcription factor 2 *(RUNX-2)* and homeobox protein MSX2. These transcription factors are critical for normal bone development, and thus their upregulation in VSMCs is indicative of a phenotypic switch. These osteocyte/chondrocyte-like VSMCs then become calcified in a process similar to bone formation. The cells lay down collagen and noncollagenous proteins in the intima or media and incorporate calcium (Ca) and phosphorus (Pi) into matrix vesicles to initiate mineralization and further grow the mineral into hydroxyapatite. The overall positive calcium and phosphorus balance of most patients undergoing dialysis feeds both the cellular transformation and the generation of matrix vesicles *(MVs)*. In addition, the extremes of bone turnover in CKD (low and high turnover or adynamic and hyperparathyroid bone, respectively) increases the available calcium and phosphorus by altering the bone content of these minerals. Ultimately, whether an artery calcifies or not depends on the strength of the army of inhibitors (Is) standing by in the circulation (fetuin-A) and in the arteries – for example, pyrophosphate *(PPi)*, matrix gla protein *(MGP)*, and osteopontin *(OP)*. (From Moe SM, Chen NX. Mechanisms of vascular calcification in chronic kidney disease. *J Am Soc Nephrol.* 2008;19:213–216. With permission.)

and in transformation (dedifferentiation), which are controlled by various transcription factors (Fig. 53.12). Expression of the osteoblast differentiation factor core binding factor α-1 (Cbfα1), now called Runx-2, has been identified in the inferior epigastric artery of adults undergoing kidney transplantation[106] and in sections from the brachial arteries of children undergoing dialysis.[107] Runx-2 is essential for normal bone development, in that Runx-2 knockout mice fail to form a skeleton.[108] Genetic techniques have confirmed that VSMCs give rise to osteochondrogenic-like cells in calcified blood vessels (as opposed to circulating cells).[109] Osteoclast-like cells can also be seen, more commonly in intimal lesions, and as in bone, they appear to arise from circulating precursors.[110]

In vitro, VSMCs upregulate Runx-2 in response to elevated Pi mediated by the type III sodium-dependent Pi cotransporters Pit-1 and Pit-2.[111] In addition, VSMCs incubated with uremic serum (pooled from anuric patients undergoing dialysis), in comparison with normal serum, express Runx-2 and its downstream protein osteopontin via a non–Pi-mediated mechanism.[112] Excess calcium can also induce mineralization in vitro, and the effects of calcium are additive to those of increased Pi.[113] Lastly, FGF-23 enhances Pi-induced vascular calcification in rat aortic rings and rat aorta VSMCs by promoting osteoblastic differentiation.[114] Numerous traditional and nontraditional cardiovascular risk factors in CKD can induce the transformation of VSMCs into osteoblast-like cells, with subsequent calcification in vitro (Fig. 53.12). Given these data, it is not surprising that arterial calcification is so common in patients with CKD.

In animal models of CKD, secondary hyperparathyroidism develops spontaneously with loss of kidney function and can be associated with vascular calcification.[48,115,116] Unfortunately, it is difficult to distinguish between the effects of Pi and PTH. However, one study found that vascular calcification developed in nephrectomized animals achieving supraphysiologic PTH

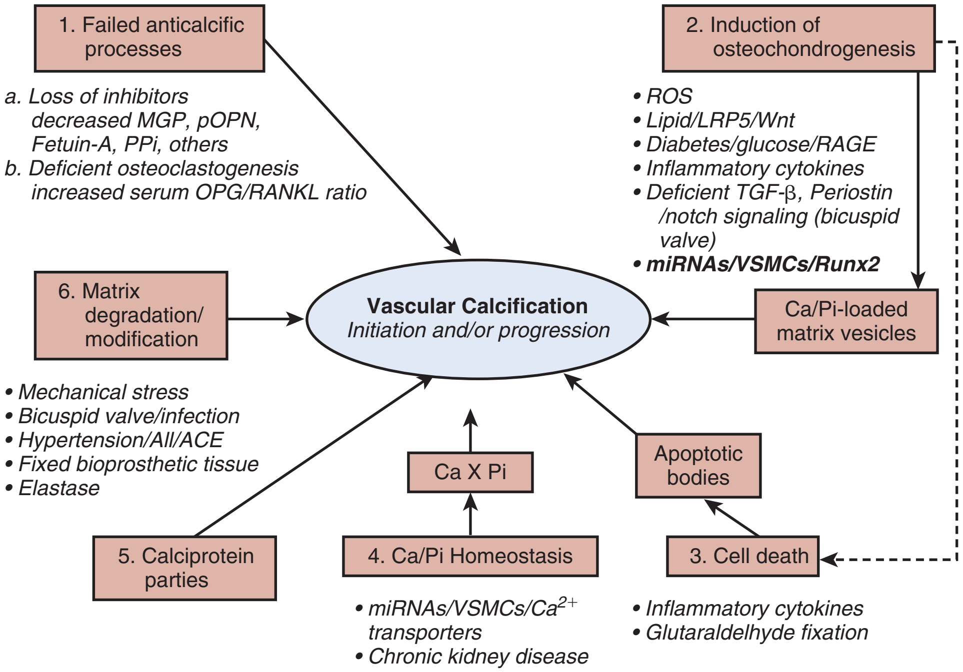


Fig. 53.12 Factors that regulate pathways involved in the pathogenesis of vascular calcification. Multiple factors regulate each step of extraskeletal calcification. *ACE,* Angiotensin-converting enzyme; *AII,* angiotensin II; *Ca,* calcium; *LRP5,* low-density lipoprotein (LDL) receptor–related protein 5; *MGP,* matrix gamma-carboxyglutamate (Gla) protein; *miRNA,* microRNA; *OPG,* osteoprotegerin; *Pi,* phosphate; *pOPN,* osteopontin; *PPi,* pyrophosphate; *RAGE,* advanced glycosylation end product receptor; *RANKL,* receptor activator of nuclear factor kappaB ligand; *ROS,* reactive oxygen species; *Runx2,* runt-related transcription factor 2; *TGF-β,* transforming growth factor β; *VSMCs,* vascular smooth muscle cells; *Wnt,* wingless-type MMTV integration site family member. (From Wu M, Rementer C, Giachelli CM. Vascular calcification: an update on mechanisms and challenges in treatment. *Calcif Tissue Int.* 2013;93:365–373. With permission.)

levels by infusion, regardless of the Pi intake.[117] In a similar study, Runx-2 was upregulated in animals in three of the four Pi/PTH intake groups—those fed normal PTH + high Pi, high PTH + high Pi, and high PTH + low Pi—indicating that both Pi and PTH may lead to Runx-2 upregulation in arteries. However, the animals given high PTH + low Pi did not demonstrate arterial calcification,[118] suggesting that Pi is needed to provide the substrate for calcification to progress.

Arterial calcification and mineralization of bone are inversely related. In animal models of excessive bone resorption, treatments aimed at decreasing bone remodeling by inhibition of osteoclast activity (i.e., bisphosphonates, calcimimetics) have been found helpful in preventing vascular calcification in some[119–121] but not all studies.[122] Correction of low-turnover bone disease also appears to improve arterial calcification in animals.[123,124] The role of vitamin D has been controversial, but data now suggest that it is only when the circulating levels of calcitriol are increased (and thus induce hypercalcemia or hyperphosphatemia) that calcification is observed.[125] Treatment studies in humans are reviewed in Chapter 63.

MATRIX VESICLES AND APOPTOSIS

In chondrocytes and osteoblasts, normal mineralization is initiated when matrix vesicles are released into the extracellular space from the cell surface via polarized budding with subsequent attachment to extracellular matrix proteins. Matrix vesicles are characterized by both their appearance as small (50–200 nm), electron-dense spherical particles on electron microscopy and the biochemical presence of calcium and Pi, alkaline phosphatase, and the membrane protein annexins. Matrix vesicles have been identified in nearly all forms of mineralization/calcification in human tissues, including bone, cartilage, tendon, calciphylaxis, and atherosclerosis. Cultured VSMCs incubated with elevated concentrations of calcium and Pi release matrix vesicles into the media, and the presence of fetuin-A (AHSG or α-2-Heremans-Schmid glycoprotein), the circulating inhibitor of mineralization (see later), decreases calcium uptake of the matrix vesicles.[126] Collagenase digestion of VSMCs has been found to lead to two populations of matrix vesicles, a secreted form in the media that had high fetuin-A and low annexin II content and could not mineralize type I collagen, and cellular matrix vesicles that had low fetuin-A and high annexin II content and could mineralize.[127] Matrix vesicles are similar to exosomes, which are known to transfer microRNAs from cell to cell and thus may play a role in calcification.[128] MicroRNAs have been found to regulate the phenotypic switch from contractile to synthetic VSMCs[129] and are involved in the regulation of arterial calcification in vitro in animal models.[128,130] These results suggest that the cellular regulation of the content of matrix vesicles may regulate the type and mineralizing capacity of the vesicle.

In addition to matrix vesicles, apoptotic bodies can induce calcification in vitro in VSMCs. Apoptotic bodies stimulated by calcium-Pi crystals of approximately 1 μm or less in diameter cause a rapid rise in intracellular calcium concentration and apoptosis, an effect triggered by lysosomal degradation.[131] Apoptosis has been identified in arterial segments with calcification obtained from children with ESKD.[107]

Activation of the DNA damage response by prelamin A accelerates arterial calcification.[132] Autophagy, a regulated process of cell survival, counteracts Pi-induced calcification by reducing matrix vesicle release.[133] By contrast, atorvastatin protects against calcification by inducing autophagy via suppression of the β-catenin pathway.[134] Thus cells appear to guard against calcification when able via a number of pathways.

INHIBITORS OF VASCULAR CALCIFICATION

Vascular calcification, although prevalent in patients with CKD and particularly in patients receiving dialysis, is not uniform. Approximately 20% (depending on the published series) of patients undergoing dialysis have no vascular calcification and continue to have no calcification on follow-up despite risk factors similar to those in patients who do have calcification. These data support the concept of calcification inhibitors. Knockout animal models have demonstrated that selective deletion of many genes leads to vascular calcification.[135] These studies imply that mineralization (or calcification) of arteries will occur, at least in some species or individuals, unless inhibited. This concept, that the regulation of calcification in blood vessels occurs principally via inhibition rather than promotion, may also be true in bone.[136] Inhibitors can be circulating or locally produced and site specific. Three inhibitors that have been well characterized in the arterial calcification of CKD are fetuin-A, matrix gamma-carboxyglutamate (Gla) protein (MGP), and OPG.

Many other inhibitors of calcification exist. In aggregate, the data discussed in this section support the diversity and abundance of naturally occurring inhibitors of calcification. Thus vascular calcification in CKD represents a state of increased calcification promoters and decreased calcification inhibitors.

Fetuin-A

Fetuin-A is a circulating inhibitor of calcification, abundant in plasma, and mainly produced by the liver in adults. The transcription and synthesis of fetuin-A are downregulated during inflammation and thus it is also a reverse acute phase reactant, like serum albumin and other hepatic proteins. Fetuin-A binds to both calcium and Pi in the serum, forming small "calciparticles" that are removed through the reticuloendothelial system. Fetuin-A inhibits the de novo formation and precipitation of the apatite precursor mineral basic calcium-Pi but does not dissolve it once the basic calcium-Pi is formed.[137] Therefore fetuin-A could be viewed as acting as a host defense to "cleanse the blood" of unwanted calcium and Pi and to prevent undesirable calcification in the circulation without causing bone demineralization. Fetuin-A has been found in matrix vesicles from VSMCs, and its presence renders the vesicles incapable of mineralization.[127,138] Fetuin-A is abundant in serum and is a major factor in the calcification propensity of serum,[139] a measure of which has been shown to reflect all-cause mortality in patients with CKD.[140]

Targeted disruption of fetuin-A leads to diffuse and profound soft tissue calcification and to arteriolar calcification of muscle, kidney, and lung but not large arteries.[141] When $Ahsg^{-/-}$ mice were crossed with $ApoE^{-/-}$ mice, the latter known to have increased cholesterol and atherosclerosis, both aorta and coronary artery calcification developed in the double-deficient $Ahsg^{-/-}/ApoE^{-/-}$ mice with high-Pi diet alone and were accelerated by CKD.[142] Thus extensive and multisite arterial calcification in this animal model required genetic predisposition to atherosclerosis (Apo $E^{-/-}$), a genetic defect in an inhibitor of mineralization ($Ahsg^{-/-}$), and hyperphosphatemia that was further accelerated by CKD. These data support the redundancy of the inhibitor system, in which multiple local regulators compensate for the absence of the circulating inhibitor fetuin-A. In patients with ESKD, lower serum fetuin-A concentrations are associated with mortality.[143,144] Serum concentrations of fetuin-A in patients undergoing dialysis were found to be inversely correlated with coronary artery calcification assessed by spiral computed tomography (CT),[145] with carotid artery plaques,[146] and, in children undergoing dialysis, arterial stiffness.[107] Fetuin-A deficiency in CKD is likely due to chronic inflammation, although gene polymorphisms may also play a role. It is also possible that there is inappropriate upregulation, leading to a relative deficiency of fetuin-A in the setting of elevated calcium and Pi.

MATRIX GAMMA-CARBOXYGLUTAMATE (GLA) PROTEIN

MGP is a vitamin K–dependent protein expressed in a number of tissues but highly expressed in arteries and bone, where it acts predominantly as a local regulator of vascular calcification. MGP knockout mice have excessive cartilage and growth plate mineralization and arterial medial calcification, resulting in early mortality.[147] In MGP-deficient mice, calcification depends on elastin fragmentation due to increased elastase production.[148] Warfarin use and/or nutritional vitamin K deficiency result in undercarboxylation of MGP and impaired function.[149] Warfarin use is also a known risk factor for calcific uremic arteriolopathy (also referred to as calciphylaxis) and can induce calcification in an animal model of CKD.[150] The administration of vitamin K can prevent calcification, although definitive trials are underway in CKD.[149] Serum concentrations of carboxylated MGP were found to be nearly undetectable in patients undergoing dialysis and in patients with atherosclerotic disease.[149] Furthermore, lower levels of carboxylated MGP were associated with coronary artery calcification, arterial stiffness, and higher serum Pi in patients undergoing dialysis.[151] Lower serum concentrations of relative carboxylated MGP as well as vitamin K deficiency were also found in 20 patients with calcific uremic arteriolopathy compared with matched controls.[152] Progression of coronary calcification was more pronounced in patients taking warfarin.[153] Supplementation of vitamin K_2 in patients undergoing dialysis can restore carboxylated MGP,[154] and studies to determine whether such supplementation reduces calcification and cardiovascular events are underway.[149]

Pyrophosphate

Another naturally occurring inhibitor of mineralization is pyrophosphate, which inhibits the formation of calcium-Pi crystals in vitro. Pyrophosphate is produced by VSMCs and inhibits arterial calcification. Pyrophosphate is inhibited by tissue-nonspecific alkaline phosphatase (TNAP), and TNAP activity is increased in calcified arteries from uremic animals[155] and patients with stage 5 CKD.[107] Pyrophosphate is also inhibited by another enzyme, ectonucleotide pyrophosphate/phosphodiesterase I (NPPI). Children deficient in NPPI have infantile arterial calcification.[156] Circulating levels of pyrophosphate are decreased in patients undergoing dialysis[157] and are negatively associated with arterial calcification in patients with CKD.[158] The intraperitoneal administration of pyrophosphate reduced vascular calcification in a rodent model of CKD.[159]

Osteoprotegerin

Osteopenia and arterial calcification develop in mice null for OPG, implying that OPG is an important direct inhibitor of vascular calcification, but it is not clear whether this development was due to abnormalities in bone[160] or a direct arterial effect. Studies in the low-density lipoprotein receptor null mice, a model of atherogenesis, demonstrated that the administration of OPG did not prevent atherosclerotic lesions but did prevent calcification of those lesions.[161] Bone marrow and vessel wall OPG reduces atherosclerosis and calcification,[162] via regulation of the procalcific effects of RANKL on VSMCs.[163] Such procalcification effects appear related to inflammation[164] and may explain mechanisms of calcification in areas of macrophage-laden atherosclerotic plaques.

INTEGRATED REGULATION OF PHOSPHORUS AND CALCIUM

The four hormones PTH, FGF-23, calcitriol, and Klotho work together to maintain normal Pi and calcium homeostasis to achieve appropriate balance in the blood and urine of these ions so as to avoid extraskeletal calcification and ensure adequate availability of these ions for bone that is growing (modeling) or remodeling. A summary of the integrated physiologic response to hyperphosphatemia is depicted in Fig. 53.13. This response is a very complex system of multiple integrated feedback loops and is easier to understand if broken into loops that regulate calcitriol, Pi, and calcium.

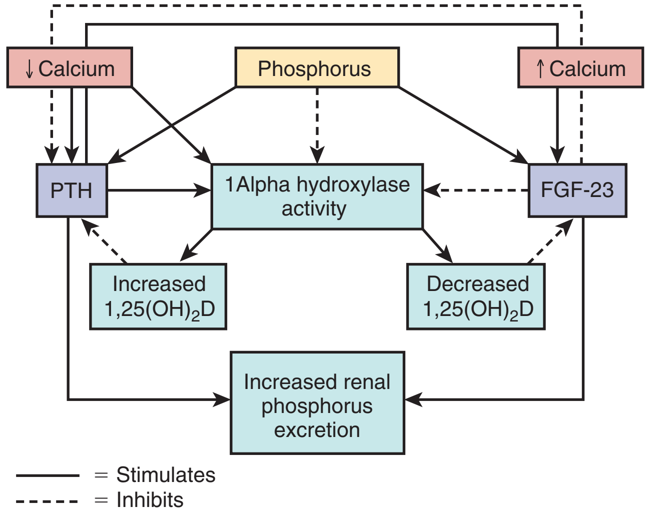


Fig. 53.13 Regulation of serum phosphorus levels. As phosphorus levels increase (or there is a long-term phosphorus load), levels of both parathyroid hormone *(PTH)* and fibroblast growth factor 23 *(FGF-23)* are increased. Both of these elevations in turn increase urinary phosphate (Pi) excretion. The two hormones differ in respect to their effects on the vitamin D axis. PTH stimulates 1α-hydroxylase activity, thereby increasing the production of calcitriol, which in turn negatively feeds back on the parathyroid gland to decrease PTH secretion. By contrast, FGF-23 inhibits 1α-hydroxylase activity, thereby decreasing the production of calcitriol, thus feeding back to stimulate further secretion of FGF-23. FGF-23 and PTH also regulate each other. Finally, low calcium levels stimulate PTH, whereas high calcium levels stimulate FGF-23. Lastly, there is some evidence that FGF-23 also inhibits PTH secretion.

PTH–FGF-23–Calcitriol Loop

PTH and FGF-23 have similar effects in stimulating Pi excretion. However, these hormones differ in their effects on the vitamin D axis. PTH stimulates CYP27B1 activity, thus increasing the production of calcitriol, which in turn negatively feeds back on the parathyroid gland to decrease PTH secretion. By contrast, FGF-23 inhibits CYP27B1 and stimulates CYP24, thereby decreasing the production of calcitriol and feeding back to limit further secretion of FGF-23—as normally calcitriol stimulates FGF-23 production.

Pi–PTH–FGF-23 Loop

As Pi levels increase (or more likely there is a long-term Pi load), both PTH and FGF-23 are increased, the latter from bone. Both the elevated PTH and FGF-23 increase urinary Pi excretion through downregulation of NaPi transporters. The effect of FGF-23 in the kidney is klotho dependent. PTH increases renal calcium reabsorption, minimizing the possibility of high calcium and Pi concentrations in urine at a time when there is a desire to increase Pi urinary excretion. PTH stimulates the secretion of FGF-23 from osteocytes, and increased FGF-23 inhibits PTH by decreasing both *PTH* gene expression and PTH secretion.[165,166]

Calcium–PTH–FGF-23 Loop

Hypocalcemia, a potent stimulator of PTH, blunts FGF-23 release.[167] The latter would therefore “remove” both the FGF-23 inhibition of PTH and the FGF-23 inhibition of calcitriol synthesis during times of hypocalcemia. This process would maximize both the PTH effects to increase renal calcium reabsorption, increase bone resorption, and enhance calcitriol stimulation of intestinal calcium absorption with the goal of normalizing serum calcium concentrations. Hypercalcemia has opposing effects: it stimulates FGF-23[168] (which reduces PTH and calcitriol synthesis) and directly inhibits calcitriol synthesis and PTH secretion. The result is decreased intestinal calcium absorption, renal reabsorption of calcium, and bone resorption.

DIAGNOSIS OF CHRONIC KIDNEY DISEASE-MINERAL AND BONE DISORDER

MEASUREMENT OF THE BIOCHEMICAL ABNORMALITIES IN CKD-MBD

The measurement of calcium and Pi were discussed earlier in this chapter. Table 53.3 summarizes currently measured biomarkers used for the diagnosis and management of CKD-MBD.

PARATHYROID HORMONE

PTH concentration in plasma or serum serves not only as an indicator of abnormal mineral metabolism in CKD-MBD but also as a noninvasive biochemical sign for the initial diagnosis of osteitis fibrosa cystica, the most common form of renal osteodystrophy in CKD-MBD. PTH measurement

Table 53.3 Biomarkers for Chronic Kidney Disease–Mineral Bone Disease

	Affected by Sample Processing	Assay Validity	Renally Excreted	Diurnal Variation	Seasonal Variation	Variation With Meals	Variation With Dialysis Time
Parathyroid hormone	Yes	No; some assays pick up fragments	No	Yes	No	No	Yes
25(OH)D (calcidiol)	No	Good (uncertain importance of differentiating D_2 from D_3)	No	No	Yes	No	No
1,25$(OH)_2$D (calcitriol)	No	Good	No	Yes	No	No	?
Fibroblast growth factor 23	No	Intact versus C terminal	No	?	?	Yes	No
Soluble α-klotho	?	Uncertain	Yes	?	?	?	?
Sclerostin	?	Uncertain, likely valid	No	?	?	?	?
Bone-specific alkaline phosphatase	No	Good	No	No	No	?	No

"Assay validity" indicates that the measurement is of the biologically active hormone or marker, not fragments.
?, Indicates insufficient data.

also can be a useful index for monitoring the evolution of renal osteodystrophy and can serve as a surrogate measure of bone turnover in patients with CKD. Although the sensitivity and specificity of PTH as a marker of bone remodeling are not ideal, it is the best marker available at the current time.[169] However, the definitive method for establishing the specific type of renal osteodystrophy in individual patients requires bone biopsy, an invasive diagnostic procedure, and access to specialized laboratory personnel and equipment capable of providing assessments of bone histology.

PTH circulates not only in the form of the intact 84–amino acid peptide but also as multiple fragments of the hormone, particularly from the middle and C-terminal regions of the PTH molecule. These PTH fragments arise from direct secretion from the parathyroid gland as well as from metabolism of PTH(1–84) by peripheral organs, especially liver and kidney. The biologically active hormone produced (PTH[1–84]) exerts its effects through the interaction of its first 34 amino acids with PTHR1. PTH(1–84) has a plasma half-life of 2 to 4 minutes. In comparison, the half-life of C-terminal fragments, which are cleared principally by the kidney, is five to ten times longer with normal kidney function and even longer in the presence of CKD. There is also a diurnal variation in the secretion of PTH and the release is oscillatory, further complicating measurement.

Assays for PTH have undergone a number of improvements over the years (Fig. 53.14). In the early 1960s, radioimmunoassays were developed for measurement of PTH. However, these assays proved to be unreliable owing to different characteristics of the antisera used and are referred to as "first-generation" assays; consequently, two-site immunometric assays (IMAs) are referred to as "second- and third-generation" assays. The typical second-generation IMAs (known as intact PTH assays) measure PTH(1–84) and other large C-terminal PTH fragments because the antibodies do not bind to amino acid 1. These assays are most commonly used in clinical practice. By contrast, third-generation assays (bioactive, whole, or bio-intact PTH assays) use capture antibody similar to that of the intact PTH assays but also use detection antibodies directed against epitopes at the extreme N-terminal end (epitopes 1–4) of the molecule, and therefore are believed to detect exclusively the biologically active PTH(1–84) (Fig. 53.14). This difference may be important because C-terminal fragments (lacking small or large portions of the N terminus) are most abundant, representing approximately 80% of circulating PTH in healthy persons and 95% in patients with CKD.[170] This finding may in part explain why elevated PTH concentrations are "normal" in CKD, yet are increased relative to values observed in patients without CKD. The second-generation intact assays are commonly used on automated platforms. Although each assay has a reasonable coefficient of variation, the standards for the commercially available assays are not uniform and the detection antibodies do not all bind at the same sites. Thus the kit-to-kit variability can be high.[171] This is the reason the Kidney Disease Improving Global Outcomes (KDIGO) guidelines recommended using the same assay every time and evaluating trends rather than targeting precise PTH values.[172]

VITAMIN D

Serum calcidiol concentrations are generally measured by immunoassay, although the gold standard for calcidiol measurement is high-performance liquid chromatography, which is not widely available clinically. Unlike in PTH assays, sample handling in calcidiol IMAs has little impact on results. However, vitamin D circulates as both D_2 and D_3, and some laboratory kits measure only D_2, others measure only D_3, and still others measure both (expressed as 25-hydroxyvitamin D). The rationale for distinguishing D_2 from D_3 is controversial, because it is unclear how differentiating the forms of vitamin D affects management or patient level outcomes. As with PTH, there is some assay-to-assay variability, which could affect the classification of insufficiency/deficiency or sufficient levels of calcidiol.[173] Fortunately, current initiatives are underway to standardize these assays.[174] The half-life of calcidiol is long and thus represents total body stores.

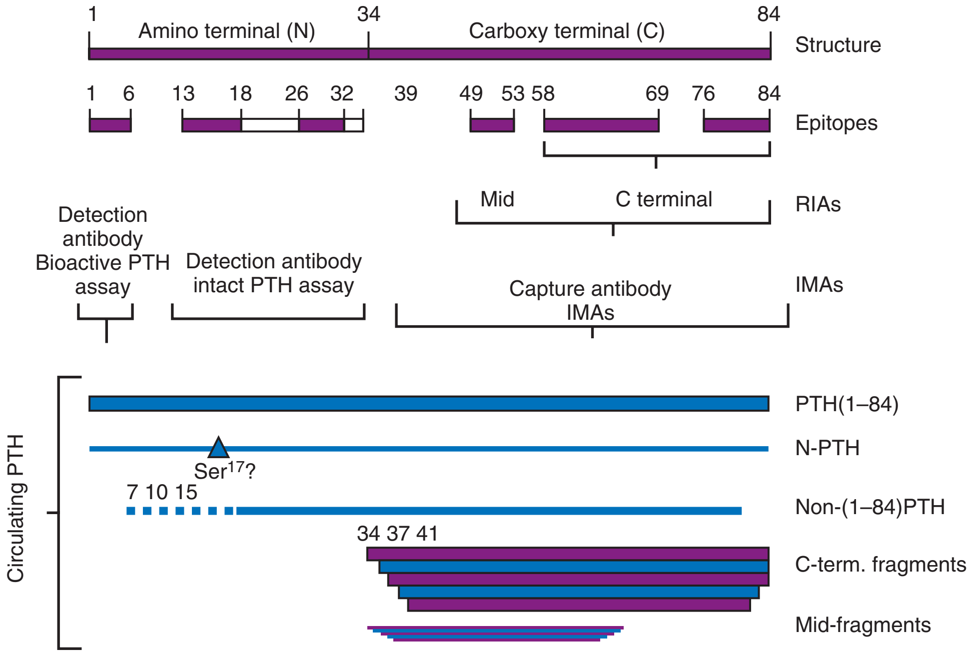


Fig. 53.14 Parathyroid hormone *(PTH)* assays. Schematic presentation of PTH(1–84) and the relationship between PTH assays, PTH assay epitopes, and PTH molecular forms detected in the circulation. The *upper panel* depicts the structure of human PTH and the epitopes detected by various PTH assays. First-generation PTH assays detect full-length PTH (1–84) in addition to PTH fragments. These assays include radioimmunoassays *(RIAs)* that use antisera specific to the amino-terminal (N-RIA), middle (MID-RIA), or carboxyl-terminal (C-RIA) regions of PTH. Second-generation "intact PTH" assays detect full-length PTH (1–84) and non(1–84)–PTH fragments. Third-generation PTH assays (bio-intact PTH) detect only full-length PTH (1–84). The *bottom panel* depicts PTH molecular forms present in the circulation. *IMA,* Immunometric assay; *N-PTH,* amino-terminal PTH; *term.,* terminal. (From Henrich LM, Rogol AD, D'Amour P, Levine MA, Hanks JB, Bruns DE. Persistent hypercalcemia after parathyroidectomy in an adolescent and effect of treatment with cinacalcet HCl. *Clin Chem.* 2006;52:2286–2293.)

By contrast, calcitriol levels are generally measured only in the setting of hypercalcemia. The half-life is comparatively short, and the assay more expensive and difficult. Interpretation also requires consideration of clinical context. In other words, a patient with stage 4 or 5 CKD and hypercalcemia may have high normal serum calcitriol, which may be distinctly abnormal given the CKD stage and serum calcium concentration, and should prompt consideration of an extrarenal source of calcitriol.

FGF-23

FGF-23 is currently measured primarily with two different assays (Fig. 53.15). The first uses two antibodies directed against the C-terminal end and thus measures the intact as well as C-terminal fragments (results are reported in relative unit/mL). The second assay uses one antibody directed against an epitope within the N-terminal region and a second antibody directed against an epitope within the C-terminal region of the molecule, and thus detects intact molecules (results are reported in picograms per milliliter). Although these two assays appear comparable in the association with clinical events at this time, they have poor agreement because of differences in FGF-23 fragment detection, antibody specificity, and calibration. Such analytical variability does not permit direct comparison of FGF-23 measurements made with different assays, a fact that probably, at least in part, accounts for some of the inconsistencies noted among observational studies.[175] FGF-23 can be detected in the urine, although at this time it is unclear how much, if any of the hormone is cleared by the kidneys.[176,177] From a clinical perspective, more data are required prior to the use of FGF-23 measurements for routine clinical management.

SOLUBLE KLOTHO

It is unclear whether the circulating or soluble α-klotho levels reflect tissue level expression of klotho. Some studies have found that low circulating levels were associated with progression of CKD,[178] but other studies have failed to confirm this finding.[179] Klotho can be detected in urine,[180] suggesting its levels may be altered by residual renal function. Different assay kits give variable results.[72,181]

SCLEROSTIN

Circulating sclerostin concentrations are elevated in CKD[182] and rise with progressive disease. However, sclerostin does not appear to be renally excreted,[183] suggesting that the rising levels reflect underlying biology. The role of sclerostin in clinical diagnosis of CKD remains exploratory, although elevated sclerostin values are associated with arterial calcification[184] and increased osteoblast number in human bone biopsy specimens.[185]

BONE-SPECIFIC ALKALINE PHOSPHATASE

Bone-specific alkaline phosphatase (BALP) is not cleared renally. BALP concentration has relatively good correlation with bone formation in CKD and may be additive to the interpretation of PTH measurements.[172] However, its concentration has limited ability as an independent measurement.[169,186]

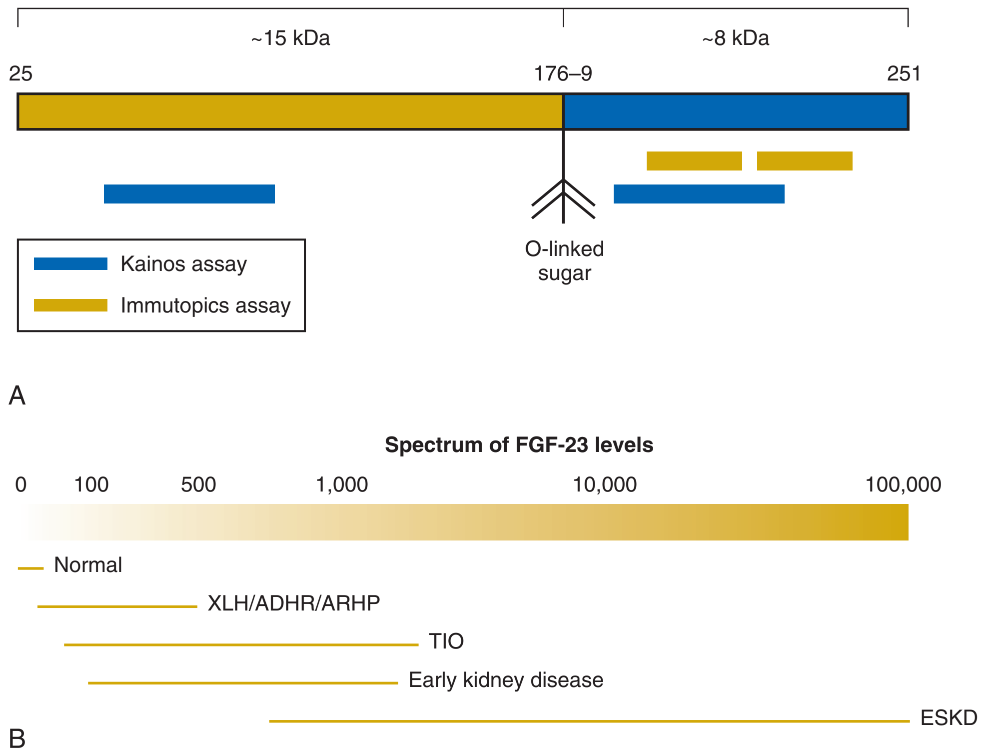


Fig. 53.15 Fibroblast growth factor 23 *(FGF-23)* assays. (A) FGF-23 O-glycosylation site and epitopes recognized by antibodies used in current assays. (B) Spectrum of serum FGF-23 levels in early chronic kidney disease and end-stage kidney disease *(ESKD)* compared with the normal reference range and levels associated with different disorders affecting FGF-23. *ADHR,* Autosomal dominant hypophosphatemic rickets; *ARHP,* autosomal recessive hypophosphatemia; *TIO,* tumor-induced osteomalacia; *XLH,* X-linked hypophosphatemia. (From Block GA, Ix JH, Ketteler M, et al. Phosphate homeostasis in CKD: report of a scientific symposium sponsored by the National Kidney Foundation. *Am J Kidney Dis.* 2013;62:457–473. With permission.)

COLLAGEN-BASED BONE BIOMARKERS

Osteoblasts secrete C- and N-terminal cleavage products of type I procollagen called secreted procollagen type IN propeptide (s-PINP) and secreted procollagen type IC propeptide (s-PICP), which are markers for bone formation. By contrast, serum C-terminal cross-linking telopeptide of type 1 collagen (s-CTX) and serum N-terminal cross-linking telopeptide of type I collagen (s-NTX) are measured as fragments of cross-links that are released when bone is resorbed. With the exception of the s-PICP, all of these markers are renally excreted, making interpretation of their measurements difficult.[187] In cross-sectional analyses, higher serum concentrations are associated with higher odds of fracture.[188]

TARTRATE-RESISTANT ACID PHOSPHATASE 5B

Tartrate-resistant acid phosphatase 5b is released by osteoclasts during bone resorption and thus may be a good marker of bone resorption.[189] However, studies relating this biomarker to bone in patients with CKD-MBD are limited.[187]

BONE BIOPSY ASSESSMENT OF BONE IN CHRONIC KIDNEY DISEASE-MINERAL AND BONE DISORDER

Abnormalities of bone quality and quantity are common in CKD-MBD (Fig. 53.16), leading to fractures and impaired growth in children. "Renal osteodystrophy" is defined as an alteration of bone morphology in patients with CKD that is quantifiable by bone histomorphometry.[1]

HISTOMORPHOMETRY IN PATIENTS WITH CKD

The clinical assessment of bone remodeling is best performed with a bone biopsy of the trabecular bone, usually at the iliac crest. The patient is given a tetracycline derivative approximately 3 to 4 weeks prior to the bone biopsy and a different tetracycline derivative 3 to 5 days prior. Tetracycline binds to hydroxyapatite and emits fluorescence, thereby serving as a label for the bone. A core of predominantly trabecular bone is collected and embedded in a plastic material, and then sectioned. The sections can be visualized with special stains under fluorescent microscopy to determine the amount of bone between administrations of the two tetracycline labels, or that formed in the interval. This dynamic parameter assessed with bone biopsy is the basis for evaluating bone turnover, which is key in discerning types of renal osteodystrophy. In addition to dynamic indices, bone biopsies can be analyzed by quantitative histomorphometry for static parameters as well. The nomenclature for these assessments has been standardized.[190]

Clinically, bone biopsies are most useful for differentiating bone turnover as well as bone volume and mineralization. However, with the advent of several new markers of bone turnover, the use of bone biopsy has been reserved primarily for the diagnosis of renal osteodystrophy and for research purposes. Sherrard and colleagues proposed a classification

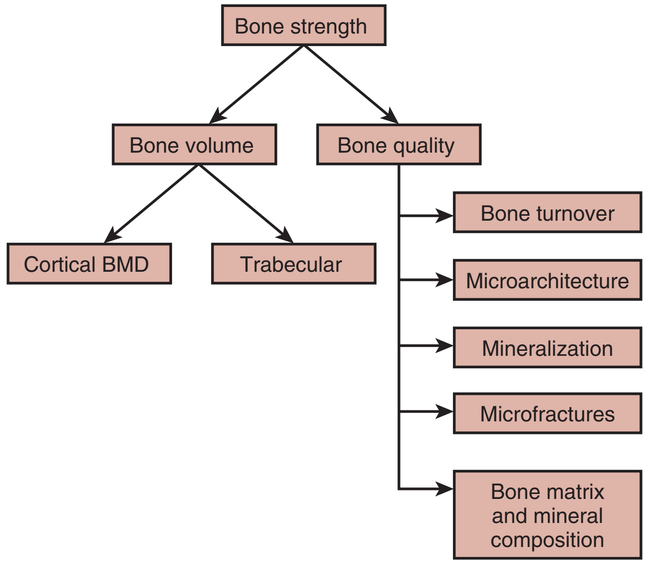


Fig. 53.16 Determinants of bone strength. Bone strength comprises both bone density and bone quality. "Bone quality" refers to bone turnover, microarchitecture, microfractures, and mineralization as well as the composition of the mineral matrix. "Trabecular microarchitecture" involves trabecular thickness, the ratio of plates and rods, and their connectivity and spacing. "Cortical microarchitecture" consists of cortical thickness, porosity, and bone size. Composition of the mineral matrix includes changes in the cross-linking of type I collagen and alterations in the size and structure of bone mineral. Bones accumulate microfractures over time even with normal physical activity. The ability to repair them affects bone quality. *BMD,* Bone mineral density. (From Moorthi R, Moe S. Recent advances in the noninvasive diagnosis of renal osteodystrophy. *Kidney Int.* 2013;84:866–894. With permission.)

system for renal osteodystrophy that utilized the parameters of osteoid (unmineralized bone) area as a percentage of total bone area and fibrosis.[191] These two static parameters, together with the dynamic bone turnover assessed by bone formation rate or activation frequency, have been used to distinguish the various forms of renal osteodystrophy over the past 30 years[191]; however, this evaluation has been replaced by the KDIGO TMV (turnover, mineralization, volume) system (discussed later).[1]

Fig. 53.17 illustrates bone histology utilizing the original classification scheme. Normal bone is illustrated in Fig. 53.17A. The histologic features of high-turnover disease (predominant hyperparathyroidism or osteitis fibrosa cystica) are characterized by increased rate of bone formation, increased bone resorption, extensive osteoclastic and osteoblastic activity, and progressive increase in endosteal peritrabecular fibrosis (Fig. 53.17B). High osteoblast activity is manifested by an increase in unmineralized bone matrix. The number of osteoclasts is also increased as well as the total resorption surface. There may be numerous dissecting cavities through which the osteoclasts tunnel into individual trabeculae. In osteitis fibrosa cystica, the alignment of strands of collagen in the bone matrix has an irregular woven pattern, unlike the normal lamellar (parallel) alignment of strands of collagen in normal bone. Although woven bone may appear to be thicker, the disorganized collagen structure may render the bone physically more vulnerable to stress.

Low-turnover (adynamic) bone disease (Fig. 53.17C) is characterized histologically by absence of cellular (osteoblast and osteoclast) activity, osteoid formation, and endosteal fibrosis. It appears to be essentially a disorder of decreased bone formation accompanied by a secondary decrease in bone mineralization. Although low-turnover disease is common in the absence of aluminum, it was initially described as a result of aluminum toxicity. Aluminum bone disease is diagnosed with special staining that demonstrates the presence of aluminum deposits at the mineralization front (Fig. 53.17D). Frequently, aluminum disease is associated with osteomalacia. Osteomalacia is characterized by an excess of unmineralized osteoid, which manifests as wide osteoid seams and a markedly decreased mineralization rate (Fig. 53.17E). The presence of increased unmineralized osteoid per se does not necessarily indicate a mineralizing defect, because larger quantities of osteoid appear in conditions associated with high rates of bone formation when mineralization lags behind the increased synthesis of matrix. Other features of osteomalacia are the absence of cellular activity and the absence of endosteal fibrosis.

Clinical Relevance

Patients with adynamic or low-turnover bone disease are at risk for fractures and vascular calcification. Common therapies for osteoporosis such as bisphosphonates and denosumab inhibit bone turnover and can result in adynamic bone disease in chronic kidney disease (CKD) and end-stage kidney disease patients. Therefore these drugs are not recommended for CKD patients unless workup reveals a high-turnover state.

"Mixed uremic osteodystrophy" is the term that has been used to describe bone biopsies that have features of secondary hyperparathyroidism together with evidence of a mineralization defect (Fig. 53.17F). There is extensive osteoclastic and osteoblastic activity and increased endosteal peritrabecular fibrosis coupled with more osteoid than expected, and tetracycline labeling uncovers a concomitant mineralization defect. Unfortunately, mixed uremic osteodystrophy, in particular, and high- and low-turnover bone diseases have been inconsistent and poorly defined.

THE SPECTRUM OF BONE HISTOMORPHOMETRY IN CKD

The prevalence of different forms of renal osteodystrophy has changed over the past decade. Whereas osteitis fibrosa cystica due to severe hyperparathyroidism had previously been the predominant lesion, the prevalence of mixed uremic osteodystrophy and adynamic bone disease has increased. However, the overall percentage of patients with high bone formation compared with low bone formation has not changed dramatically over the last 20 to 30 years, although osteomalacia has been essentially "replaced" by adynamic bone disease. There are differences in prevalence of mixed uremic osteodystrophy in patients not yet undergoing dialysis, which appears to depend on the level of GFR and the country in which the study was performed.[192] Two large analyses of

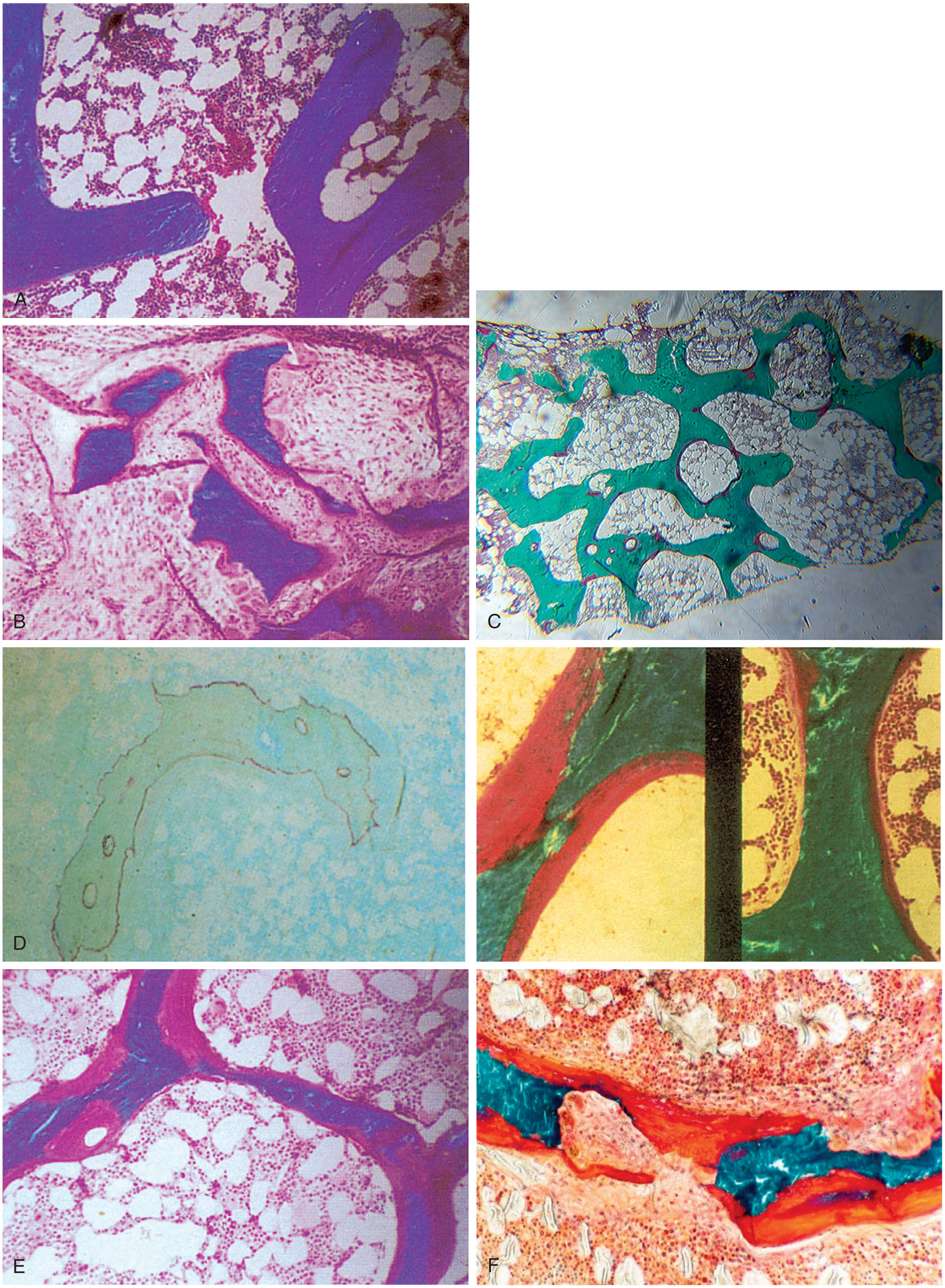


Fig. 53.17 Bone histology. (A) Normal bone. (B) Hyperparathyroid bone (increased osteoclast and osteoblasts and fibrosis). (C) Adynamic bone (no cellular activity and no osteoid). (D) Aluminum bone disease (left aluminum staining at mineralization front) and right two panels show accumulation of osteoid *(orange-red stain).* (E) Osteomalacia (increased unmineralized osteoid in *pink/red*). (F) Mixed uremic osteodystrophy presence of increased osteoid *(orange red)* indicating mineralization defect, and increased osteoclast activity. (A, B, D *(right)*, and E, courtesy S.L. Teitelbaum, MD; D *(left)*, courtesy D. J. Sherrard, MD.)

patients undergoing long-term dialysis revealed a relatively high incidence of low bone turnover; one study of 489 biopsy specimens in predominantly Caucasian patients revealed low turnover in 59%,[169] whereas in another study of 630 patients, low-turnover disease was noted in 62% of Caucasian but only in 32% of African-American patients.[193] The prevalence of mineralization defect or osteomalacia was relatively low at only 3%.[193] Thus these data demonstrate that histologic abnormalities of bone begin very early in the course of CKD and that differences in bone turnover may be based on racial differences.

By contrast, low-turnover bone disease has diverse pathophysiology. In the 1980s, aluminum-induced osteomalacia was common. The potential toxicity of aluminum was initially recognized by Alfrey, who identified a fatal neurologic syndrome in patients receiving dialysis consisting of dyspraxia, seizures, and electroencephalographic abnormalities in association with high brain aluminum levels on autopsy.[194] The source of aluminum in these severe cases was believed to be elevated concentrations in dialysate water. Subsequently, aluminum-containing phosphate binders were also identified as a source. The additional symptoms of fractures, myopathy, and microcytic anemia were described several years after the initial reports of the neurologic syndrome.[194,195] Fortunately, exposure to aluminum is limited in the modern era, and the incidence of aluminum bone disease is relatively rare. However, the diagnosis of aluminum-induced bone disease can be difficult, because aluminum toxicity is due to tissue burden, not serum concentrations. Thus if aluminum bone disease is suspected, bone biopsy remains the gold standard for making the diagnosis.[195,196]

In adynamic bone disease, there is a paucity of cells with resultant low bone turnover (Fig. 53.17C). Unlike in osteomalacia, in adynamic bone there is no increase in osteoid or unmineralized bone. The lack of bone cell activity led to the initial description of the disease as "aplastic" bone disease. Early investigators believed that the disease was due to aluminum, but it was later identified in the absence of aluminum. The etiology of adynamic bone disease is likely multifactorial, and major contributory factors include diabetes mellitus, aging, and malnutrition.[197]

Proposed pathophysiologic mechanisms of low bone turnover are listed in Table 53.4. Increases in both sclerostin and dkk-1, which are soluble inhibitors of wnt signaling that inhibit osteoblastic bone formation, likely play a role in development of adynamic bone disease.[97,198,199] Circulating fragments of PTH (7–84–amino acid fragments) may also be antagonists to PTH,[200] resulting in an effective resistance to 1–84 amino acid at the level of bone. There is evidence that markedly elevated concentrations of FGF-23 may be associated with decreased osteoblastic activity.[201] Furthermore, abnormal regulation of cell differentiation in the presence of renal failure may explain, in part, the relative paucity of cells in adynamic bone, although this possibility remains to be proven. In rats, the administration of bone morphogenetic protein-7 can restore normal cell function, supporting that a failure of normal cell differentiation, likely due to a number of causes, may be critical.[202] Although most patients with low-turnover bone disease are asymptomatic, they are at increased risk of fracture owing to impaired remodeling[201,203,204] and at risk of vascular calcification because of the inability of bone to buffer a sudden calcium load.[201,205,206]

Table 53.4 Causes and Proposed Mechanisms of Decreased Bone Formation in Patients With Chronic Kidney Disease

	Mechanism of Decreased Osteoblast Activity
Low serum calcitriol	↓ Osteoblast differentiation ↓ Osteoblast life span
Metabolic acidosis	↓ Calcitriol production ↓ Collagen synthesis
High serum phosphate	↓ Calcitriol production
Calcium loading/ hypercalcemia	↓ Calcitriol production and ↑calcitriol degradation (mediated by calcium-sensing receptor)
High serum interleukins 1 and 6, tumor necrosis factor	↓ Osteoblast life span
Low serum insulinlike growth factor-I (IGF-I) activity	↓ IGF-I and IGF binding protein 5 (IGFBP5) levels ↑ Inhibitory IGFBP (2, 4, 6) levels ↓ Osteoblast life span
Sclerostin	↓ Wnt/β-catenin signaling ↓ Osteoblastic activity
Dickkopf-1 (dkk-1)	↓ Wnt/β-catenin signaling ↓ Osteoblastic activity
Malnutrition, proteinuria	↓ IGF-I; 25-hydroxyvitamin D levels
Diabetes	↓ 25-Hydroxyvitamin D and calcitriol levels ↑ Advanced glycation end products (AGEs) ↓ Osteoblast life span
Age-related	↑ AGEs ↓ Osteoblast life span
Hypogonadal	
Women (↓ estrogen and ↑ sex hormone–binding globulin [SHBG])	↓ Osteoblast life span
Men (↓ testosterone and ↑SHBG)	↓ Osteoblast life span
Uremic toxins (uric acid)	↓ Calcitriol production ↓ Vitamin D receptor activity ↓ Osteoblast proliferation
Aluminum toxicity	↓ Osteoblast activity

↓, Decreased; ↑, increased.

TMV CLASSIFICATION

As previously noted, KDIGO recommends that the definition of renal osteodystrophy be limited to describing the alterations of bone morphology in patients with CKD and is one measure of the skeletal component of the systemic disorder of CKD-MBD that can be quantifiable by histomorphometry.[1] Historically, "renal osteodystrophy" often included disorders of bone mineral metabolism in addition to histomorphometric changes in bone. As our understanding of bone biology progresses, there is greater appreciation of the diverse physiologic processes leading to similar bone biopsy findings. In addition, information on bone volume as an independent

Table 53.5 Turnover, Mineralization, Volume Classification System for Renal Osteodystrophy

Turnover	Mineralization	Volume
Low		Low
	Normal	
Normal		Normal
	Abnormal	
High		High

From Moe S, Drüeke T, Cunningham J, et al. Definition, evaluation, and classification of renal osteodystrophy: a position statement from Kidney Disease: Improving Global Outcomes (KDIGO). *Kidney Int.* 2006;69:1945–1953.

parameter is available.[207] Thus the previous classification was updated by KDIGO to use three key histologic descriptors – bone turnover, mineralization, and volume (TMV system), with any combination of each of the descriptors possible in a given specimen (Table 53.5).[1] The TMV classification scheme provides a clinically relevant description of the underlying bone pathology as assessed by histomorphometry, which in turn helps define the pathophysiology and thereby guide therapy.

Turnover reflects the rate of skeletal remodeling, which is normally the coupled process of bone resorption and bone formation. Turnover is assessed with histomorphometry by dynamic measurements of osteoblast function using double-tetracycline labeling, as previously described. Bone formation rate and activation frequency represent acceptable parameters for assessing bone turnover. Bone turnover is affected mainly by hormones, cytokines, mechanical stimuli, and growth factors that influence the recruitment, differentiation, and activity of osteoclasts and osteoblasts. It is important to clarify that although bone formation rate is frequently similar to bone resorption rate, which cannot be measured directly, it is not always so. Imbalance in these processes can affect bone volume. For example, if resorption exceeds formation, negative bone balance and decreased bone volume result.

Mineralization reflects how well bone collagen becomes calcified during the formation phase of skeletal remodeling. It is assessed with histomorphometry by static measurements of osteoid volume and osteoid thickness and by dynamic, tetracycline-based measurements of mineralization lag time and osteoid maturation time. Causes of impaired mineralization include inadequate vitamin D nutrition, mineral (calcium or Pi) deficiency, acidosis, and aluminum toxicity.

Volume indicates the amount of bone per unit volume of tissue. It is assessed with histomorphometry by static measurements of bone volume in cancellous bone. Determinants of bone volume include age, sex, race, genetic factors, nutrition, endocrine disorders, mechanical stimuli, toxicities, neurologic function, vascular supply, growth factors, and cytokines.

The KDIGO classification is consistent with the historically used classification system[191] but provides more information on parameters other than turnover. Two large-scale analyses utilizing the updated TMV system suggested that the newer TMV classification system provides clinically relevant information.[169,193] Low bone volume and low bone turnover are more common than heretofore appreciated, whereas defective mineralization is relatively rare in adult patients with ESKD.

Clinically, serum PTH concentration is used as a surrogate biomarker to predict bone turnover. However, as detailed earlier, studies evaluating the ability of the serum concentration of (intact) PTH to predict low- and/or high-turnover bone diseases have been disappointing. The ability to reliably predict the presence of high-turnover bone disease is poor until serum PTH concentrations exceed 500 pg/mL.[169,193] Primarily on the basis of earlier studies utilizing the Allegro intact PTH assay, the Kidney Disease Outcomes Quality Initiative (KDOQI) guidelines recommend a target-intact PTH level of 150 to 300 pg/mL.[195] Unfortunately, studies that correlate intact PTH with bone histology were performed with an assay no longer available. Use of currently available assays have shown that the intact PTH value between 150 and 300 pg/mL is not predictive of underlying bone histology.[208] However, an analysis of 610 biopsy specimens found that although the intact PTH value could not predict underlying bone histology, it was able to discriminate low-turnover from non–low-turnover bone diseases and high-turnover from non–high-turnover bone diseases.[169]

NONINVASIVE ASSESSMENT OF BONE

DUAL-ENERGY X-RAY ABSORPTIOMETRY

Dual-energy X-ray absorptiometry (DXA) measures areal bone mineral density (aBMD or simply BMD) in g/cm^2 using minimal radiation and rapid scan times. BMD assessment by DXA has good reproducibility (<1%–2% coefficient variation) and reliable reference ranges for age, sex, and race. In the general population, BMD measured by DXA can be used clinically to define osteoporosis and is an accepted surrogate endpoint after prospective studies demonstrated an age-dependent predictive value of DXA for fractures.[209] However, the discordance between changes in DXA findings and the efficacy of drugs at preventing fracture has led to appreciation of the importance of bone "quality," which is not assessed by DXA. This observation has forced the use of fractures as endpoints for approval of new therapeutics for the treatment of osteoporosis.

The KDIGO CKD-MBD guideline recommends DXA to assess fracture risk in patients with stage 1 through early stage 3 CKD, as long as biochemical testing does not suggest CKD-MBD.[172] However, for patients with CKD stages 3b through 5, the guideline did not recommend DXA owing to the lack of definitive data demonstrating that DXA predicts fracture in CKD-MBD. A meta-analysis was performed to determine whether DXA measurements at the femoral neck, spine, and radius were associated with spine and/or nonspine fractures in patients undergoing dialysis.[210] On the basis of results of six studies with 683 patients, lower BMD in the spine and distal radius, but not in the femoral neck, was associated with fractures. Although BMD may have been lower in patients with CKD and a history of fracture, there is considerable overlap in BMD such that BMD provides poor fracture discrimination in individuals. In addition, for lumbar spine assessment of BMD by DXA, existing aortic calcification may confound the measurement. Since the publication of KDIGO guidelines, studies have demonstrated fracture prediction value of DXA in CKD subjects at least equivalent to that in the general population.[211,212] These data therefore support that DXA can predict fracture risk in patients with CKD. However, DXA cannot make a specific diagnosis as to why there is low BMD. Unlike patients with normal kidney function and low BMD by DXA being classified as having osteoporosis, patients with CKD and low bone density should not be routinely treated with antiosteoporosis therapies.[213] DXA is an inexpensive and widely available technique that can be easily standardized among sites, so it may be a good tool in longitudinal CKD research studies for the serial assessment of BMD in response to interventions. Unfortunately, to date, treatment studies utilizing DXA as an endpoint in patients with CKD are limited.

QUANTITATIVE COMPUTERIZED TOMOGRAPHY TECHNIQUES

Quantitative CT (QCT) allows three-dimensional imaging of cross sections of the central and axial skeleton to provide spatial or volumetric BMD (vBMD). It also allows distinction between cortical and trabecular compartments. In CKD, QCT measures of trabecular bone density at the spine have been correlated with trabecular bone volume histomorphometry.[214] Peripheral QCT (pQCT) avoids the large dose of ionizing radiation exposure for patients by focusing on the tibia and distal radius, and in one study, was associated with fracture.[215] Although derived from a single study, these results are in line with expected associations between loss of bone quality and fracture risk in patients undergoing dialysis.

High-resolution peripheral computerized tomography (HRpQCT) has greater resolution than pQCT and allows evaluation of trabecular microarchitecture (bone volume fraction, trabecular thickness, separation, and number). HRpQCT of the radius and tibia was found to discriminate among patients with CKD with and without fracture.[216,217] However, the discriminatory power of the various HRpQCT parameters to individually discriminate between those with and without fractures by receiver operating curve analyses was less than 0.75. When patients with the longest duration of CKD were considered, the area under the receiver operating curve improved to more than 0.8 for multiple parameters, including radial cortical thickness, radial total vBMD, and cortical vBMD. Interestingly, aBMD by DXA of the ultra-distal radius performed similarly in its ability to discriminate prevalent fractures in this subpopulation with the longest duration of kidney disease.[218] In another study, areal BMD by DXA at the ultra-distal radius was superior to HRpQCT measures for fracture discrimination.[219] Therefore high-resolution CT techniques do provide assessment of bone architecture, but their use at this time is limited to research centers and their additive value over distal radius DXA has not yet been proven.

MICROCOMPUTED TOMOGRAPHY AND MICROMAGNETIC RESONANCE IMAGING

Resolution of microtechniques is as low as 8 mm, compared with about 100 μm for HRpQCT, thus providing spatial resolution almost that of an actual bone biopsy. Micro-CT was used to assess the effects of severe high-turnover renal osteodystrophy on trabecular and cortical architecture in growing rats. The technique demonstrated irregular trabecular thickening and loss of trabecular connectivity in the femoral neck and greater endocortical porosity in the femoral shaft, consistent with biopsy findings in high-turnover osteodystrophy.[220] Unfortunately, this technique is currently limited to in vitro studies. In an evaluation of 17 patients undergoing hemodialysis who had secondary hyperparathyroidism, micromagnetic resonance imaging (micro-MRI) demonstrated disruptions of the distal tibial trabecular network.[221] There have not been subsequent confirmatory studies or studies comparing this technique with other imaging methods in the ability to predict fractures in CKD.

ASSESSMENT OF VASCULAR CALCIFICATION

Arterial calcification can be detected in humans through a number of techniques. Plain radiographs can be used to assess the presence or absence, and thus prevalence, of vascular calcification. Scoring methods utilizing the number of aorta segments calcified along the lumbar spine can also allow reproducible quantification during longitudinal follow-up,[222] although the sensitivity is less than with CT-based imaging. Although some distinction can be made between medial and intimal calcification on plain radiographs,[223] the reproducibility among multiple research sites of this method for differentiation of calcification type has not been evaluated. Ultrasonography of the carotid arteries can be used to assess intimal–medial thickness, which correlates well with atherosclerosis and cardiovascular events. In a later study, intravascular ultrasonography of the coronary arteries was able to detect atherosclerotic lesions and calcification. This technique, although invasive, can detect circumferential lesions, and external remodeling of atherosclerosis—lesions that invade the internal elastic artery into the medial layer rather than protrude into the vessel lumen. The latter luminal lesion is all that is detected on angiography, leading some to question the use of angiography as the "gold standard."[224] Finally, technologic advances have led to ultra-fast CT scans—electron beam CT (EBCT) and multislice CT—that use electrocardiography gating to allow imaging only in diastole, thus avoiding motion artifact of the heart. These techniques have allowed reproducible quantification of coronary artery and aorta calcification and therefore serve as excellent intermediate endpoints for CKD-MBD interventions that might alter progression of vascular calcification. Unfortunately, these techniques do not allow differentiation of medial from intimal calcification, and the radiation dose with their use is large.

A study in each of 140 prevalent patients undergoing hemodialysis compared a lateral radiograph of the lumbar abdominal aorta, an echocardiogram, and measurement of pulse pressure with EBCT results.[225] Calcification of the abdominal aorta was scored as 0 through 24 divided into tertiles; echocardiograms were graded as 0 through 2 for absence or presence of calcification of the mitral and aortic valves; and pulse pressure was divided in quartiles. The researchers found that the likelihood ratio (95% confidence interval [CI]) of coronary artery calcification score on EBCT of 100 or higher was 1.79 (1.09–2.96) for calcification of either valve on echocardiography and 7.50 (2.89–19.5) for participants with a lateral abdominal radiographic score greater or equal to 7. Verbeke and colleagues confirmed this predictive value of lateral abdominal radiographs in more than 1000 patients receiving dialysis.[226]

On the basis of observational studies, calcification detected on all of these imaging studies is associated with mortality in patients with CKD. Bellasi and Raggi analyzed multiple such studies and calculated the positive predictive value to be between 19% and 52% and the negative predictive value to be between 68% and 100%.[227] The addition of the clinical variables age and dialysis vintage to abdominal calcification, termed the "cardiovascular calcification index," was linearly associated with death, with a 12% increase in hazard ratio for each point increase in the cardiovascular calcification index.[228]

CLINICAL CONSEQUENCES OF THE ABNORMALITIES IN CHRONIC KIDNEY DISEASE-MINERAL AND BONE DISORDER

BIOCHEMICAL ABNORMALITIES

PHOSPHORUS AND CALCIUM

Epidemiologic data suggest that serum Pi levels above the normal range are associated with increases in morbidity and all-cause and cardiovascular mortality in patients with CKD. These studies differ in their sample size, analyses, and chosen reference ranges, and most evaluate only patients undergoing dialysis. In patients with CKD stages 2 to 5, higher levels of serum Pi, even within the normal range, have been associated with increased risk of all-cause or cardiovascular mortality in one study[229] but not in other studies.[230,231] However, in patients undergoing dialysis, nearly all studies demonstrate an association of elevated Pi with mortality, although there are slight differences in the inflection point or range at which Pi becomes significantly associated with increased all-cause mortality. Furthermore, several retrospective studies have demonstrated that treatment of patients undergoing dialysis with Pi binders is associated with a 20% to 40% lower risk of death.[232–234] However, there are no prospective studies that have demonstrated a survival benefit of lowering serum phosphate concentrations, and there is no specific target serum phosphate that has been identified as conferring a clinically relevant benefit. In patients with CKD stages 3 to 5, there are no data to support an increased risk of mortality or fracture with elevated serum calcium concentrations. However, as with Pi, there are several studies in patients receiving dialysis demonstrating higher mortality with higher serum calcium concentration and in some studies, among patients with very low serum calcium concentrations, as the latter may reflect more severe vitamin D deficiency and secondary hyperparathyroidism.

PARATHYROID HORMONE

Although PTH has long been considered a surrogate marker for bone disease, it also has systemic effects, including effects on endothelial, cardiac, and skeletal health. As with other biochemical measures of CKD-MBD, observational studies have found an association of all-cause mortality with various levels of PTH, with inflection points ranging from more than 400 to 600 pg/mL. On the basis of these observational data and the assay limitations, the KDIGO guidelines consider serum concentrations of intact PTH within two and nine times the upper limit of normal for the PTH assay (<130 and >585 pg/mL for most kits with an upper normal limit of 65 pg/mL) as a reasonable target range, and that values within that range should be interpreted by evaluation of trends. Intervention should occur if the trends are consistently going up or down.[172] However, it is important to recognize that there are no randomized clinical trials demonstrating that treatment to achieve specific PTH concentrations results in improved outcomes, with the exception of the Evaluation Of Cinacalcet Hydrochloride (HCl) Therapy to Lower CardioVascular Events (EVOLVE) trial discussed in Chapter 63.

A Cochrane meta-analysis found that serum Pi and calcium concentrations, but not PTH, were associated with cardiovascular and all-cause mortality.[235]

COMBINATION OF ABNORMALITIES OF CALCIUM, PHOSPHORUS, AND PARATHYROID HORMONE

The relationship of calcium, Pi, and PTH with outcomes is further complicated by the clinical reality that these laboratory parameters do not respond in isolation from one another, but rather change depending on the levels of other parameters and treatments. For example, Stevens and colleagues assessed various biochemical combinations and found that the relative risk for mortality was highest when serum concentrations of calcium and Pi were elevated and the serum PTH concentration was low and mortality was lowest when serum concentrations of calcium and Pi were normal and the serum PTH concentration was high.[236] In addition, results were modified by vintage (time since initiation of dialysis). A Dialysis Outcomes and Practice Patterns Study (DOPPS) trial also evaluated combinations of serum parameters of mineral metabolism and reached slightly different conclusions.[237] The DOPPS researchers found that in the setting of an elevated serum PTH (>300 pg/mL), hypercalcemia (>10 mg/dL) was associated with higher mortality risk even with normal serum Pi. Block and associates, analyzing average values over 4 months from more than 26,000 patients undergoing dialysis, categorized them according to whether serum concentrations of calcium, Pi, and PTH were below, within, or above target ranges. In only 20% of patients were all three variables within target ranges. Those patients with high PTH and high serum calcium had consistently higher mortality.[238] It is important to remember that abnormalities in mineral metabolism are common. Ultimately, we need quality evidence (i.e., from prospective randomized controlled trials using approaches that target specific biochemical endpoints) to determine whether these treatments—despite having been commonly administered in some cases for decades—are truly safe and efficacious.

FIBROBLAST GROWTH FACTOR 23

Serum concentrations of FGF-23 (>100 RU/mL) were observed in 70% and 100% of patients with estimated GFRs of 50 and less than 20 mL/min/1.73 m^2, respectively, in the Chronic Renal Insufficiency Cohort (CRIC) study.[73] Serum FGF-23 concentrations continue to rise throughout progressive CKD, presumably to maintain hemostasis vis-à-vis Pi and calcium. In CKD, elevations of serum FGF-23 are associated with death,[239] progression of CKD,[239,240] left ventricular hypertrophy,[78,241,242] and cardiovascular events,[243,244] independent of serum concentrations of Pi and PTH. Once patients reach levels of kidney function poor enough to require dialysis, serum concentrations of FGF-23 can be 1000-fold greater, and these have been associated with poorer survival in incident patients in some studies[245] but not others.[246,247] In prevalent patients receiving dialysis, serum FGF-23 was also associated with mortality[248] and directly correlated with left ventricular mass index.[247] In a secondary analysis of nearly 2000 hemodialysis patients treated with cinacalcet, a reduction of serum FGF-23 concentrations by 30% or more was associated with reduced risks of death and cardiovascular events.[249]

NUTRITIONAL VITAMIN D DEFICIENCY

Although traditionally the term "vitamin D" has been used to indicate the active metabolite, calcitriol, the correct use of the term is for the precursor molecule, calcidiol or 25(OH) D. In the general population, serum calcidiol concentrations are accepted as the standard measures of nutritional uptake because they correlate best with end-organ effects. The conversion of vitamin D_2 and vitamin D_3 to 25(OH)D by CYP27A1 is essentially unregulated, and therefore levels of 25(OH)D are a reliable indicator of the vitamin D status of a given individual. However, what constitutes adequate stores remains a controversy. Although there is no absolute level of calcidiol that defines deficiency, a level less than 10 ng/mL (25 nmol/L) is typically used, because it is associated with rickets in children and osteomalacia in adults. The term "vitamin D insufficiency" has been used to describe less severe calcidiol-deficient states. Although controversial, the typical range of "insufficient" calcidiol levels is 10 to 30 ng/mL (25–75 nmol/L). In contrast to the severity of bone disease observed with deficiency, insufficiency is associated with elevated PTH and osteoporosis.[250] Unfortunately, despite supplementation of calciferol in various foods such as milk, calcidiol insufficiency is relatively common in the general population, particularly in some at-risk groups such as African-Americans, hospitalized patients, nursing home residents, and people living in northern climates.[251] The Institute of Medicine convened a panel whose report, released in November 2010, recommended that serum vitamin D concentrations of 20 ng/mL are adequate and that the adult-recommended dietary allowance is 600 IU/day, with an upper level intake of 4000 IU/day.[252] The Endocrine Society has subsequently recommended a vitamin D level higher than 30 ng/mL, further suggesting that 40 to 60 ng/mL is ideal and that up to 100 ng/mL is safe.[253] Either vitamin D_2 or vitamin D_3 can be used for vitamin D supplementation. Although there has been controversy regarding the use of D_3 or D_2 for achieving and maintaining higher serum vitamin D levels, most analyses have found them to be equally effective in raising and maintaining serum vitamin D levels in patients without CKD.[254]

The importance of vitamin D in health and disease is suggested by a number of observational studies, with the effects of local cellular (autocrine) conversion thought responsible for the purported nonendocrine benefits. Vitamin D deficiency (<20 ng/mL) is likely to be an important etiologic factor in the pathogenesis of many chronic diseases, including autoimmune diseases (e.g., multiple sclerosis, type 1 diabetes), inflammatory bowel disease (e.g., Crohn disease), infections, immune deficiency, cardiovascular diseases (e.g., hypertension, heart failure, sudden cardiac death), cancer (e.g., colon cancer, breast cancer, non-Hodgkin lymphoma), and neurocognitive disorders (e.g., Alzheimer disease).[253,255–260] The association between serum vitamin D concentrations and all-cause mortality was inverse and nonlinear, with increased risk evident at serum vitamin D concentrations below 30 ng/mL. Vitamin D deficiency was also associated with significantly higher rates of death due to cancer and respiratory diseases.[260] However, the only prospective study of vitamin D versus placebo was unable to demonstrate that vitamin D supplementation resulted in a lower incidence of invasive cancer or cardiovascular events.[261]

The results of randomized trials testing the safety and efficacy of vitamin D supplementation on fractures and falls have been mixed. Among the reasons put forth to explain mixed results is the concomitant use of calcium in most studies, and another is the wide range in dose of vitamin D utilized. According to the latest update of the Cochrane Review analysis of vitamin D and vitamin D analogs for prevention of fractures in postmenopausal women and older men, vitamin D alone is unlikely to prevent fractures in the doses and formulations tested so far. However, supplementation of vitamin D and calcium may prevent hip or any type of fracture. In this meta-analysis, vitamin D and calcium in combination resulted in a small but significant increase in gastrointestinal symptoms and kidney disease. The researchers found no evidence that calcium and vitamin D increased (or decreased) the risk of death.[262]

Vitamin D deficiency (<20 ng/mL) and insufficiency (<30 ng/mL) are common in patients with CKD, with only 29% and 17% of those with stage 3 and 4 disease, respectively, having sufficient levels in one study.[263] This prevalence of vitamin D deficiency and insufficiency in an ambulatory CKD population is similar to that in non-CKD nursing home residents, hospitalized patients, and elderly women with hip fractures. However, frank vitamin D deficiency (<10 mg/mL) was observed in 14% and 26% of patients with stage 3 and 4 CKD, respectively.[263] Several other investigators have also found widespread vitamin D insufficiency in patients with CKD, although not all investigators have identified a graded relation with CKD stage.[263–266]

Evidence from epidemiologic studies suggests that vitamin D deficiency and supplementation of vitamin D are associated with survival in patients with CKD regardless of dialysis status, although these analyses are heavily confounded, prohibiting causal inference.[267–272] A meta-analysis of prospective studies demonstrated a significant decrease, 14%, in mortality risk for every 10-ng/mL increase in serum vitamin D concentration.[273] Furthermore, the association of vitamin D supplementation and survival appears to be independent of changes in serum calcium, Pi, and PTH concentrations. There are several plausible mechanisms that could explain the connection between vitamin D deficiency and insufficiency and mortality; for example, serum vitamin D concentrations are inversely correlated with multiple cardiovascular risk factors, including plasma renin activity, blood pressure, left ventricular mass, markers of inflammation, markers of insulin resistance and type 2 diabetes mellitus, and albuminuria.[274–278]

In animal studies, activated vitamin D inhibits renin production.[279,280] Several trials in patients with CKD reported that the use of an active vitamin D analog, paricalcitol, resulted in a significant reduction in proteinuria.[281–283] Evaluation of patients with nondialysis-requiring CKD found an inverse correlation between serum vitamin D concentrations and brachial artery flow–mediated dilation as a marker of endothelial dysfunction.[284] Other studies have demonstrated that patients undergoing hemodialysis who have received cholecalciferol supplements have lower inflammatory parameters, as indicated by higher serum albumin and lower C-reactive protein and interleukin-6 concentrations, as well as improved cardiac function, as reflected by lower brain natriuretic peptide and left ventricular mass index.[285,286] In a prospective placebo-controlled trial in patients who had not started dialysis, paricalcitol failed to reduce left ventricular mass after 48 weeks in patients with mild-to-moderate left ventricular hypertrophy. There was, however, a decrease in hospitalizations, a secondary endpoint.[287] The lack of effect may have been a function of the relatively high dose of paricalcitol, which may have increased serum FGF-23 concentrations, as multiple studies have shown a direct link between FGF-23 and left ventricular hypertrophy.[78,241,287,288] Despite multiple observational studies and several clinical trials, additional controlled trials are required to determine the safety and efficacy of (nutritional and active) vitamin D supplementation in patients with CKD.

FRACTURE

The incidence of fractures was reported to be significantly higher in patients with nondialysis-requiring CKD compared with persons with normal or near-normal kidney function, although lower than the risk of fractures in patients receiving dialysis,[172,289–291] suggesting that kidney disease is associated with progressive bone deterioration. However, not all studies have confirmed increased rates of hip, wrist, and vertebral fractures in patients with CKD independent of age and sex.[292] More recent studies have suggested that fracture rates in ESKD patients are declining compared with historical data.[293]

In patients undergoing dialysis, the studies are consistent and demonstrate a markedly higher incidence of hip fracture among all age groups.[203,294] Patients undergoing hemodialysis may have a 50% higher rate of hip fractures than patients undergoing peritoneal dialysis.[295] Furthermore, hip fractures in patients on dialysis were associated with a doubling of mortality observed after hip fracture in patients not on dialysis.[203,296] Depending on the study, 10% to 52% of prevalent patients undergoing dialysis have had fractures.[172] Unfortunately, most of these studies are of cross-sectional design or were cohort studies with limited follow-up.

Hyperparathyroidism is known to cause abnormal bone remodeling, especially of cortical bone. Serial studies using HRpQCT show a rapid decline in cortical bone in patients on dialysis.[297] In prevalent patients undergoing dialysis who have secondary hyperparathyroidism, clinical fractures are predominantly cortical in location.[298] However, the association of PTH with fracture has been inconsistent,[172] perhaps because both high and low PTH values are associated with fractures. Other risk factors for hip fracture include older age, female sex, low serum albumin, prior kidney transplantation, and peripheral vascular disease.[188,299–301] Falls are also more common in patients with CKD and predispose to fracture. Falls in patients undergoing dialysis may be related to peripheral vascular disease,[294] low muscle strength, impaired neuromuscular function,[302] and the administration of psychoactive medications.[300]

In summary, the abnormalities of bone, in terms of both quality and quantity, variable biochemical abnormalities, and multiple comorbid conditions likely all contribute to the pathogenesis of fractures in ESKD. This complexity also may limit generalizability of therapies routinely used in the general population to patients with the disease.

EXTRASKELETAL CALCIFICATION

Arterial calcification has a multitude of clinical manifestations. Intimal calcification (in association with atherosclerotic disease) can lead to myocardial infarction from stenosis and acute thrombus or to ischemia in both coronary and peripheral arteries. Medial calcification (or circumferential calcification) can lead to arterial stiffening, with reduced compliance of the artery and an inability to appropriately dilate in the setting of increased stress. In the coronary arteries, similar symptoms of ischemia can develop, and in theory could lead to arrhythmias and sudden death. In the larger arteries such as the aorta, calcification can lead to increased pulse wave velocity and elevated pulse pressure and is commonly associated with systolic hypertension in the elderly, a known risk factor for cardiovascular disease in the general population. In addition, the premature return of wave reflections during systole (instead of diastole) can lead to altered coronary perfusion, especially in the setting of left ventricular hypertrophy. Lastly, calcification of the arterioles of the skin and other organs can lead to localized infarction and ischemia, including ischemic bowel and calcific uremic arteriolopathy (calciphylaxis).

VASCULAR CALCIFICATION IN PATIENTS WITH CHRONIC KIDNEY DISEASE

The high prevalence of vascular calcification in patients with CKD is not a new observation. In 1979, Ibels and colleagues demonstrated that both renal and internal iliac arteries of patients undergoing renal transplantation had greater atherogenic/intimal disease and increased calcification (detected by biochemical methods) than transplant donors.[303] In addition, the medial layer was thicker and more calcified in uremic patients than in the donors. A study comparing histologic changes in coronary arteries from patients undergoing dialysis at autopsy were compared with nondialysis, age-matched patients who had died from a cardiac event found similar magnitudes of atherosclerotic plaque burden and intimal thickness, but with more calcification in the patients who had received dialysis.[304] In addition, morphometry of the arteries demonstrated increased medial thickening. When these same investigators evaluated more distal segments of the coronary arteries, they found medial calcification.[305] Studies of the inferior epigastric artery of patients undergoing kidney transplantation demonstrated calcification in 31% of patients, with histology demonstrating isolated medial calcification without evidence of atherosclerotic or intimal changes, and both medial and intimal calcification in the same artery without disruption of the internal elastic lamina.[306] Thus there is histologic evidence of (1) increased arterial calcification in coronary, renal, and iliac arteries of patients who undergo dialysis than in those who do not and (2) the presence of both intimal and medial calcification in patients undergoing dialysis, and this can occur independently of each other, suggesting unique inciting factors. As noted previously, it is likely that calcification of plaque (atherosclerosis) and calcification of medial layer occur by similar mechanisms. However, the initiating event is clearly different.

Braun and colleagues first demonstrated that coronary artery calcification as detected by EBCT increased with advancing age in patients on dialysis and that the calcification scores were twofold to fivefold higher in patients on dialysis than in age-matched persons with normal kidney function and angiographically proven coronary artery disease.[307] Nearly 50% to 60% of patients starting hemodialysis have evidence of coronary artery calcification,[308] with higher prevalence in patients with diabetes.[309] The prevalence of detectable coronary or peripheral artery calcification is 51% to 93% in prevalent patients undergoing dialysis and 47% to 83% in patients with CKD stages 3 through 5. Valvular calcification is similarly common, occurring in 20% to 47% of patients on dialysis.[172] Studies that have evaluated the natural history of coronary calcification demonstrate that once present, the course is progressive. By contrast, many patients without calcification remain free of calcification for several years.[172,310,311] The presence of calcification in the coronary arteries, valves, and peripheral arteries is associated with mortality.[227] Given the high prevalence of calcification, it is currently uncertain whether routine imaging offers clinical utility or whether it can help stratify patients whose calcification will and will not respond to specific therapies.

There is an inverse relationship between bone mineralization and vascular calcification in patients on dialysis. There are several different, not mutually exclusive, mechanisms by which disturbances in the bone metabolism may cause or accelerate vascular calcification.[307,312] Interestingly, it appears to be low-turnover bone disease that leads to greatest risk of vascular calcification. London and colleagues evaluated patients on hemodialysis who underwent assessment of vascular calcification by ultrasonography with semiquantitative scoring and by bone biopsy with histomorphometry. Those patients with lowest bone formation rates and decreased osteoblast surfaces had the most prominent degree of peripheral artery calcification, and this relation held true in patients with and without previous parathyroidectomy.[313] Another study found that patients with low-turnover bone disease on biopsy were more likely to have progression of coronary artery calcification over 1 year as assessed by serial multislice CT than those with high-turnover bone disease.[208] Abnormalities in cortical and trabecular bone on HRpQCT were associated with coronary artery calcification.[314] The likely mechanism relates to a reduced ability to mineralize bone in ESKD, especially in the setting of low-turnover bone disease, leading to dystrophic calcification in the setting of calcium (and vitamin D) intake; actively remodeling bone, by contrast, is able to mineralize, as demonstrated by Kurz and associates.[205] In contrast to evidence in low-turnover bone disease, data are less robust as to the effects of high-turnover bone disease on vascular calcification, perhaps because the treatments for high-turnover bone disease confound the effects of calcification.

CHRONIC KIDNEY DISEASE-MINERAL AND BONE DISORDER IN KIDNEY TRANSPLANT RECIPIENTS

Bone and mineral disorders are universal complications in patients with CKD prior to transplantation. Ideally, all of these complications would be improved with successful kidney transplantation. Unfortunately, in many cases, kidney transplantation achieves milder forms of CKD (as opposed to normal kidney function), and thus kidney transplant recipients still suffer from disorders associated with CKD-MBD. In addition, disorders of mineral metabolism occur following successful transplantation, including the effects of medications (steroids and calcineurin inhibitors), persistence of underlying disorders (hyperparathyroidism and vitamin D deficiency), development of hyperphosphaturia with hypophosphatemia (especially with persistent hyperparathyroidism).

BIOCHEMICAL CHANGES AFTER TRANSPLANTATION

Biochemical changes of mineral metabolism are common in patients after kidney transplantation.[315–318] Decreased calcidiol levels are common in kidney transplant recipients, with reported prevalence of up to 81%.[316] The causes are multifactorial, and may include nutritional deficiency, malabsorption, and decreased sun exposure. However, the improvement in kidney function results in an increase in CYP27B1 activity, resulting in greater conversion of calcidiol to calcitriol, which subsequently causes a gradual decrease in PTH concentrations during the first 3 to 6 months after transplantation. However, persistent hyperparathyroidism exists in about 33% and 20% of patients at 6 and 12 months, respectively.[319] Serum FGF-23 concentrations decrease immediately after transplantation, although they remain inappropriately elevated for the prevailing serum Pi concentration. By about 3 months after transplantation, serum FGF-23 concentrations are reduced by 89%, and at 12 months FGF-23 concentrations appear to be similar to those in patients without transplantation who have comparable stages of CKD.[316] The elevated FGF-23 inhibits CYP27B1 and contributes to the low calcitriol concentrations observed in the early post-transplantation period, likely contributing to the persistent hyperparathyroidism seen after transplantation. Thus elevated FGF-23 and PTH concentrations combined with low calcitriol concentrations likely contribute to hypophosphatemia observed in many transplant recipients.[319,320]

Serum calcium concentrations typically drop very early after transplantation, as a result of cessation of therapy with calcitriol and its analogs and/or calcium-containing phosphate binders. After that initial drop, however, calcium concentrations progressively increase, with hypercalcemia developing in a substantial number of patients during the first to third month after transplantation. Hypercalcemia can be transient, resolving by 6 to 8 months in some cases but lasting for many years in others. Hypercalcemia is typically associated with hyperparathyroidism and results from enhanced renal tubular reabsorption of calcium under the influence of PTH in the functioning allograft, the effects of calcitriol on the gastrointestinal absorption of calcium, and, potentially, a direct effect of PTH in causing calcium efflux from bone. Low bone turnover with relatively low PTH may also result in hypercalcemia because the bone is not able to act as a buffer for the circulating calcium pool.[315,316]

BONE CHANGES AFTER TRANSPLANTATION

Disorders of mineral metabolism and associated bone disease lead to fractures following transplantation, which occur after up to 44% of successful kidney transplants.[315] This fracture risk appears to be less than that observed with other solid organ transplants.[315] The combination of kidney–pancreas transplantation raises fracture risk above that associated with kidney transplantation alone.[321,322] In a retrospective analysis of 68,814 patients, fracture developed in 22.5% within 5 years.[323] In a study of first-time kidney transplant recipients between 1997 and 2010, the incidence of hip fracture was 3.8 per 1000 person-years. The risk of fracture in 2010 was 0.56 (95% CI, 0.47–0.77) in comparison with that in 1997, indicating a decrease over the 13-year period, presumably due to changes in immunosuppression therapy.[324] Factors associated with higher fracture risk during the first 5 years included female sex, age older than 45 years, Caucasian race, recipient of a deceased donor kidney, increased human leukocyte antigen mismatches, diabetes, pretransplantation dialysis, steroid use, and an aggressive induction regimen.[323] Importantly, even with the use of steroid-sparing regimens, total (all sites) fractures have occurred in up to 50% of kidney transplant recipients at 6 years.[324,325] By contrast, a Canadian study evaluated 4821 kidney transplant recipients and found that the 10-year cumulative incidence of hip fracture was 1.7%.[326]

Studies evaluating bone histology in recipients of kidney transplants are limited. The main alteration is an uncoupling of bone remodeling, which results in a decrease of bone formation with persistent bone resorption and net bone loss. In a small cohort of young patients who underwent preemptive transplantation and were treated predominantly with corticosteroids, bone biopsies revealed a mineralization defect as early as 6 months after transplantation.[327] Another study evaluating the early histologic changes after transplantation revealed that osteoid volume, osteoid thickness, osteoid resorption surface, and osteoclast surface were elevated above the normal range before transplantation and remained increased approximately 35 days afterward. However, osteoid and osteoblast surfaces, which also were increased before transplantation, were significantly decreased approximately 35 days after transplantation.[328] Both bone formation and mineralization were also inhibited after transplantation. An important observation was that although none of the pretransplantation biopsy specimens showed evidence of apoptosis, 45% of posttransplantation specimens showed significant apoptosis after an average of only 35 days. Thus early posttransplantation apoptosis and a decrease in osteoblast number and osteoblast surface play a role in the pathogenesis of posttransplantation bone disease that may be related directly to the use of glucocorticoids. The few evaluations that examined the longer-term effect of transplantation on bone histology generally demonstrate low bone turnover and histology consistent with adynamic bone disease.[329,330] Lehmann and colleagues performed a retrospective study of 57 patients after approximately 54 months after kidney transplantation.[330] Seven histologic subgroups were identified: normal, osteitis fibrosa cystica, mild osteitis fibrosa, mixed uremic bone disease, osteomalacia, adynamic bone disease, and osteoporosis. Given the limited number of patients in the study, no significant correlations between the histological subgroups and PTH or alkaline phosphatase levels were observed. However, this study did show a variable degree of histologic abnormalities following transplantation that could not have been predicted by biochemical testing and confirmed that preexisting bone disease lesions persist in the posttransplantation period.

The effects of other immunosuppressive agents on bone histology have also been examined. Bone biopsies performed approximately 10 years after transplantation in patients whose treatment included cyclosporine monotherapy, azathioprine and prednisone, or triple therapy revealed no differences among the immunosuppressant regimens.[331,332] A subgroup analysis of 21 patients with normal serum PTH concentrations showed that cyclosporine monotherapy was associated with a more pronounced decrease in bone thickness than the other regimens.[332] In addition, the cyclosporine group showed a lower trabecular appositional rate than the azathioprine and prednisone group. Multiple regression analysis showed that sex and time after transplantation were the most significant factors predicting bone volume and mineralizing surface. Predictive factors for eroded bone surface and osteoclast number included age and time on dialysis before transplantation.[331,332]

Initial studies examining BMD by DXA showed severe and rapid bone loss after kidney transplantation.[327] However, later studies have not found the dramatic decrease initially described, perhaps because of the current practice of using lower steroid dosages or steroid-free immunosuppression regimens.[315,333] The use of corticosteroids is the major determinant of low BMD, because these agents impair calcium absorption from the gastrointestinal tract and inhibit bone cell recruitment and function. Cueto-Manzano and associates also evaluated BMD as a function of immunosuppressive therapy.[331,332] In comparison with age- and sex-matched controls, none of the treated groups showed a significant reduction in BMD in the lumbar spine or femoral neck. However, in comparison with young normal controls, osteopenia was detected in the femoral neck, except in premenopausal women. There was no significant difference in BMD related to immunosuppressant regimen, sex, or menopausal status.

Although reductions in BMD have been associated with an increased fracture rate in postmenopausal women and in men treated with glucocorticoids, as well as in heart or liver transplant recipients, very little data support its role in predicting fractures after kidney transplantation. In the one study that demonstrated low BMD to be predictive of fracture risk, 283 kidney transplant recipients with varying degrees of kidney function underwent 670 examinations of bone density at the hip. An absolute BMD value less than 0.9 g/cm^2 was associated with an increased risk of fracture.[334] A study evaluating patients with early steroid withdrawal after kidney transplantation found preservation of BMD by DXA but continued loss at the ultra-distal radius. The latter finding was confirmed by HRpQCT to be predominantly cortical loss, which was associated with persistent hyperparathyroidism.[335] As indicated earlier, abnormal bone turnover may alter the predictive value of BMD testing in patients with CKD, and thus preexisting rate of bone turnover likely affects the assessment and outcomes of bone disease in transplant recipients.

VASCULAR CALCIFICATION CHANGES AFTER TRANSPLANTATION

Most transplant recipients enter their posttransplant course with a preexisting burden of vascular calcification; as in patients with CKD, vascular calcification strongly predicts cardiovascular events and all-cause mortality. Unfortunately there are few studies of the nature and consequences of vascular calcification in kidney transplant recipients. Although preexisting calcifications progress after transplantation, they appear to do so at a much slower rate than prior to transplantation.[336,337] A review of 13 clinical studies of the role of vascular calcifications in kidney transplant recipients shows that as in the CKD population, the associations of traditional and nontraditional risk factors with outcomes are variable in transplant recipients.[338] There is a strong association between baseline coronary artery calcification score and progression of coronary artery calcification. A significant improvement in secondary hyperparathyroidism after transplantation is associated with slowing the progression of coronary artery calcification. Independent risk factors of coronary artery calcification in kidney transplant recipients include low serum concentrations of calcidiol, fetuin-A, and MGP. Although diabetes is a risk factor for the presence of coronary artery calcification in transplant recipients, it has not been independently associated with progression of coronary artery calcification. Data on the effects of immunosuppressive drugs as factors in progression are few and inconclusive; however, it does appear that mycophenolate mofetil may have a beneficial effect because its antiproliferative action may inhibit smooth muscle cell proliferation, thus having a favorable effect on endothelial cell activity.[338] In the general population and in patients with ESKD, there is an inverse relation between coronary artery calcification and bone density. This association has not been shown in kidney or kidney–pancreas transplant recipients.[339]

SUMMARY

In summary, some abnormalities of CKD-MBD persist after kidney transplantation, and other conditions unique to transplantation frequently develop. With changing immunosuppressive protocols and persistence of CKD, treatment of these conditions can be particularly challenging. Recent studies have highlighted the importance of management of CKD-MBD before transplantation. Additional studies are required to determine the safety and efficacy of established and novel therapies in patients with CKD and ESKD and after kidney transplantation.[317,318,340]

Complete reference list available at ExpertConsult.com.

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BOARD REVIEW QUESTIONS

1. A 56-year-old woman with ESRD on hemodialysis is evaluated for painful nodules on the abdomen which have been present for the past month. Medications include warfarin, metoprolol, atorvastatin, renal vitamin, calcium acetate. She is receiving doxercalciferol and darbopoietin iv with dialysis. Physical examination reveals an overweight female with several hard, subcutaneous nodules approximately 1 cm in diameter in the lower abdomen; there is one open wound 1.5 cm in diameter with surrounding erythema. Laboratory studies reveal the following results: calcium 9.7 mg/dL; phosphate 6.0 mg/dL; PTH 550 pg/mL. She undergoes a punch biopsy of the wound which shows findings consistent with calcific uremic arteriolopathy. Which of the following therapeutic approaches would you recommend?
   a. discontinue warfarin
   b. start sodium thiosulfate with dialysis
   c. start vitamin K supplemetation
   d. a and b
   e. a, b, and c

   Answer: d

   Rationale: Calciphylaxis or calcific uremic arteriolopathy is a rare but devastating complication in CKD and ESRD, with mortality approaching 50%. Epidemiological studies have linked warfarin use with calciphylaxis, and a potential mechanism is related to the inhibition by warfarin of the calcification inhibitor matrix Gla-protein (MGP). MGP is gamma-carboxylated similar to factors in the coagulation cascade, and higher amounts of undercarboxylated MGP are correlated with vascular calcification. While vitamin K supplementation in theory should rescue the defect in MGP, no data is yet available from ongoing studies on the efficacy of vitamin K supplementation in reducing or reversing vascular calcification. In epidemiological studies and case reports, sodium thiosulfate was shown to be at times useful in treatment of calciphylaxis, although the exact mechanism by which it reduces extra-osseous calcification is not known.

2. A 60-year-old female patient 6 months post kidney transplant is evaluated in routine follow-up. Physical exam is unremarkable. Laboratory studies reveal the following: creatinine 1.0 g/dL, calcium 10.8 mg/dL; phosphate 2.8 mg/dL; PTH 150 pg/mL. Which of the following statements about her condition is true?
   a. Hyperparathyroidism is uncommon following a successful kidney transplant
   b. She should be referred to an endocrine surgeon for parathyroidectomy
   c. Cinacalcet may offer effective therapy for her hyperparathyroidism
   d. Post-transplant hypercalcemia is not associated with increased mortality or graft loss
   e. She should have FGF23 levels checked given borderline low phosphate levels

   Answer: c

   Rationale: Post renal transplant hyperparathyroidism is commonly observed, although usually resolves by 6 to 12 months post-transplant. The associated hypercalcemia has been associated with increased mortality and graft loss in epidemiological studies, and thus control of the hypercalcemia is indicated. While parathyroidectomy can be used for refractory cases or those that fail medical management, cinacalcet is currently a reasonable option, especially early in the post-transplant period. FGF23 levels may be increased in post-transplant patients and contribute to the hypophosphatemia, although the primary endocrine abnormality in this case is hyperparathyroidism, not hyperphosphatonism.

3. Which one of the following statements about phosphate control in patients with CKD not on dialysis is correct?
   a. phosphate binders effectively lower phosphate levels in CKD patients not on dialysis
   b. phosphate binders have been shown to decrease vascular calcifications and mortality in placebo controlled trials
   c. phosphate levels drawn during an office visit accurately reflect total body phosphate stores
   d. vegetarian protein sources have less bioavailable phosphate than animal protein sources
   e. The optimal phosphate level of a patient with CKD is less than 4 mg/dL

   Answer: d

   Rationale: Whereas epidemiological studies have shown that higher phosphate levels are associated with increased mortality, no prospective placebo-controlled study has shown that lowering phosphate levels to a particular target lowers clinical outcomes in the CKD or ESRD populations. Few studies have been done in CKD patients, and they have not shown consistent decrease in phosphate levels, partly because circulating phosphate levels do not adequately reflect total body stores. Phosphate in plant sources is less bioavailable because some of it is in phytate which humans cannot digest, and small studies using vegetarian protein sources compared with animal product protein sources have shown effect not only of phosphate levels but also on FGF23.

4. Which of the following statements are true concerning the integrated regulation of calcium and phosphorus?
   a. Both PTH and FGF 23 stimulate the activation of 25 hydroxycholecalciferol to 1,25 dihydroxycholecalciferol to enhance gastrointestinal absorption of calcium and phosphate.
   b. Both hypocalcemia and PTH stimulate the release of FGF23.
   c. Both FGF23 and 1,25 dihydroxycholecalciferol increase the production of PTH
   d. Increased phosphate load in CKD results in increases of both PTH and FGF23 which downregulate the renal sodium phosphate transporters.
   e. b and d

   Answer: d

   Rationale: Hyperphosphatemia, or more specifically increased phosphate loading cause increase in both PTH and FGF-23, the latter from bone. Both the elevated PTH and FGF-23 increase urinary phosphate excretion through downregulation of NaPi transporters. The effect of FGF-23 in the kidney is klotho dependent. PTH also causes increase in calcium reabsorption, thus decreasing urinary calcium. PTH stimulates the conversion of 25 hydroxycholecalciferol to 1,25 dihydroxycholecalciferol, by stimulating CYP27B1 activity, whereas FGF23 inhibits CYP27B1 and stimulates CYP24, thereby decreasing 1,25 dihydroxycholecalciferol levels. Hypocalcemia is a major stimulus for PTH release, however, it has an inhibitory effect on FGF23, whereas hypercalcemia will stimulate FGF23. Both FGF23 and 1,25 dihydroxycholecalciferol inhibit the release of PTH.